Credit for materials: I am extremely grateful the many researchers who have made their slides and course materials available. With particular thanks to Prof. Michalis Vazirgiannis (EP), Prof. David Fouhey (NYU), Prof. Olga Russakovsky (Princeton) and Prof. Vikram V. Ramaswamy (Princeton). This lecture is also inspired by the course CS231n: Deep Learning for Computer Vision from Stanford.

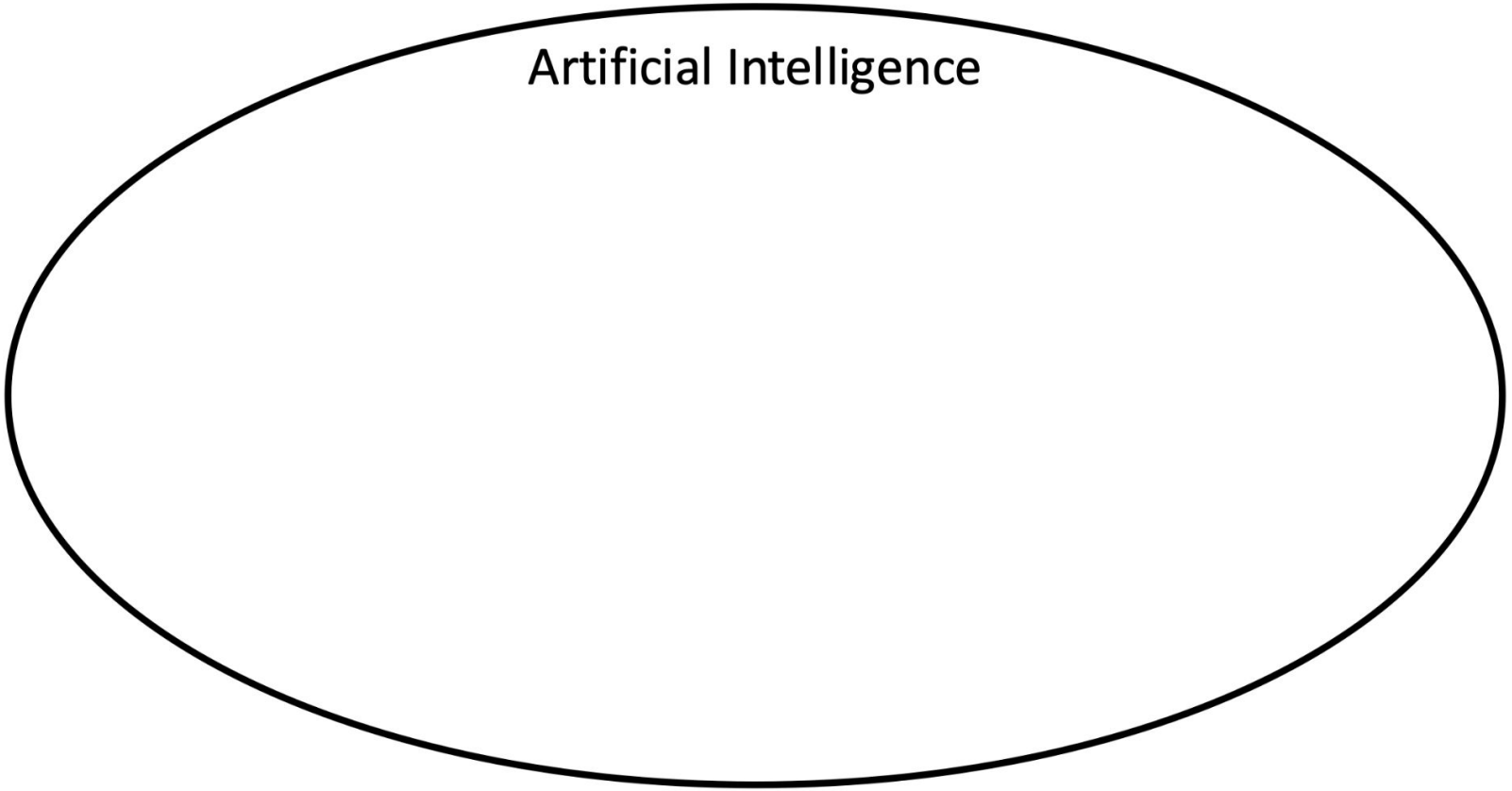
Introduction to Image Processing and Computer Vision in Deep Learning

CSC_51054_EP: Deep Learning
Ecole Polytechnique, France

Lecture Instructor: Prof. Ye ZHU
Monge Assistant Professor
https://l-yezhu.github.io/

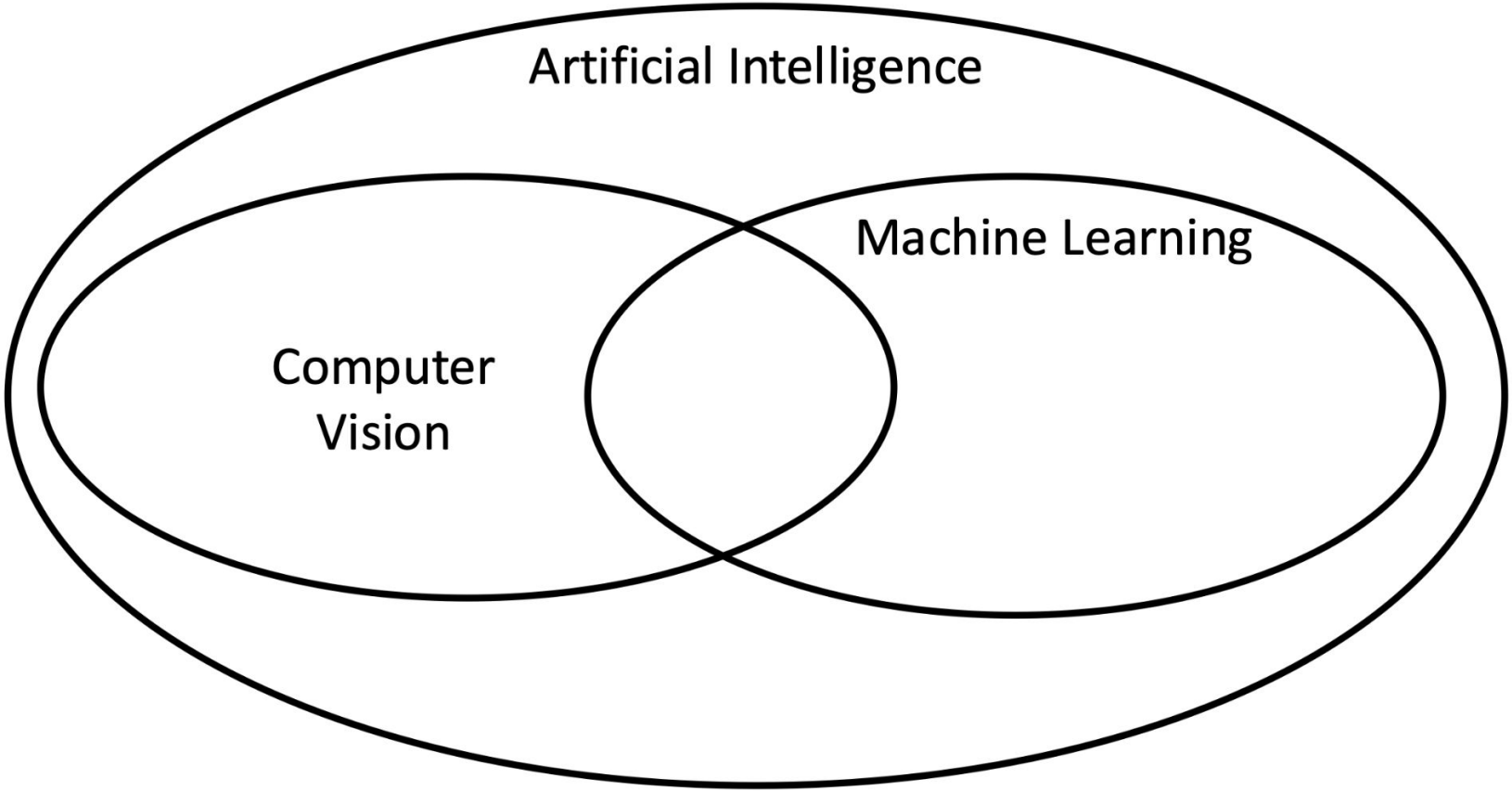
November, 2025

Where are we in the world of AI?



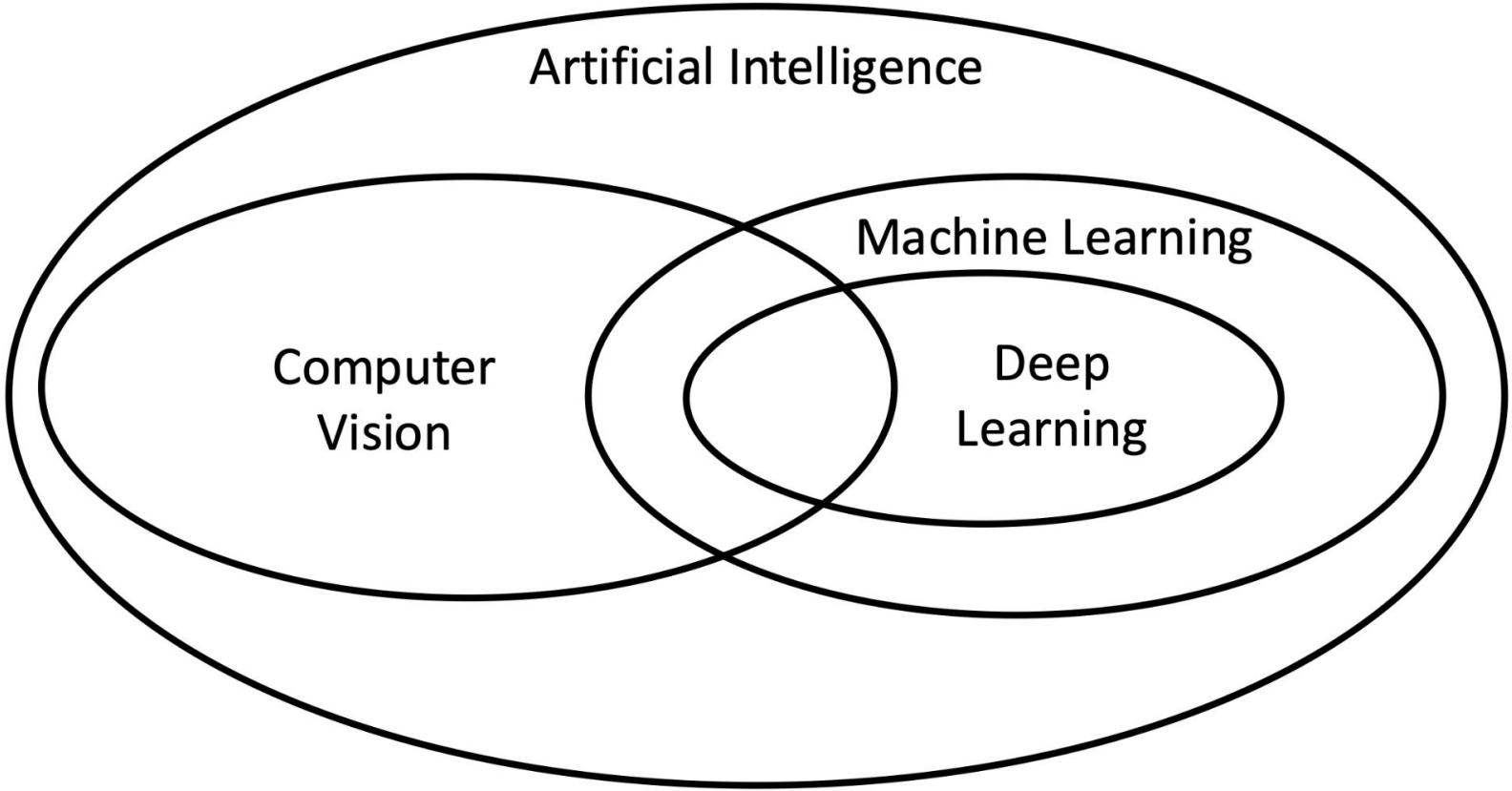
Slide inspiration: Justin Johnson

Where are we in the world of AI?



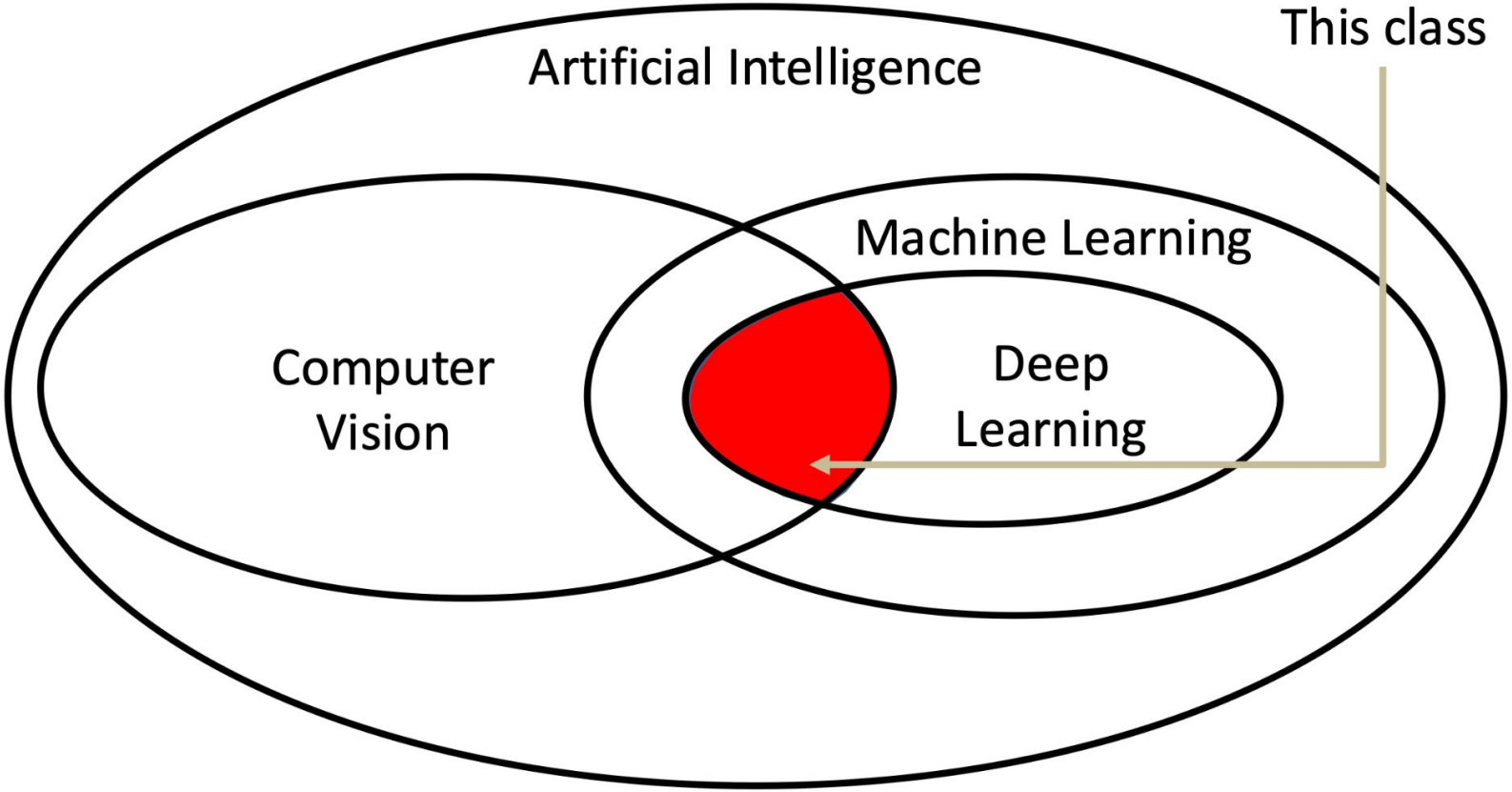
Slide inspiration: Justin Johnson

Where are we in the world of AI?



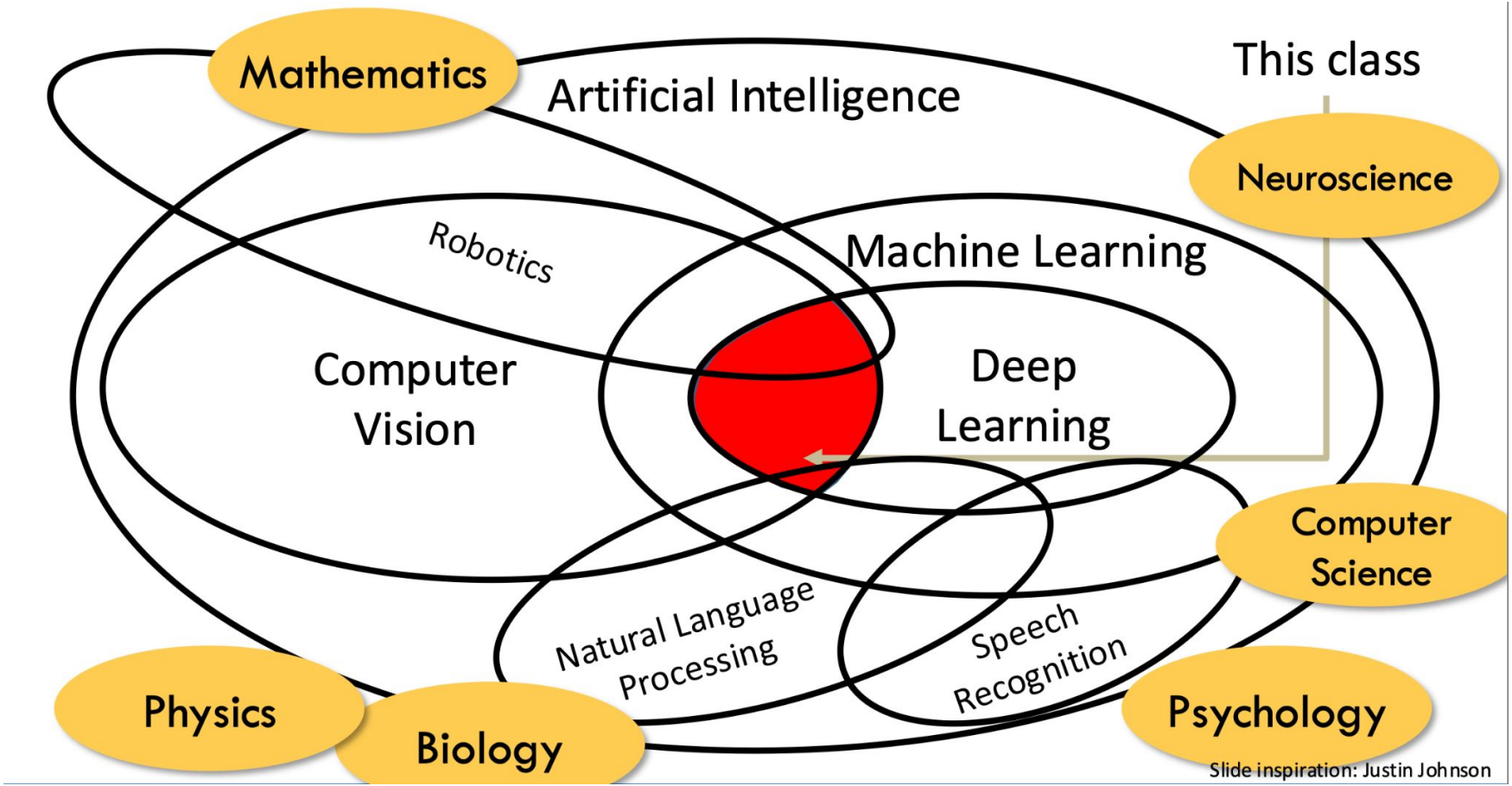
Slide inspiration: Justin Johnson

Where are we in the world of AI?



Slide inspiration: Justin Johnson

Where are we in the world of AI?



Today's agenda

- A brief history of computer vision and deep learning
- Visual perception and understanding: What makes a good visual representation? *(Lab to follow after the lecture!)*
- Classic image processing techniques
- Image classification and beyond
- Object detection and segmentation
- Other more advanced topics on visual generation and multimodal learning

Today’s agenda

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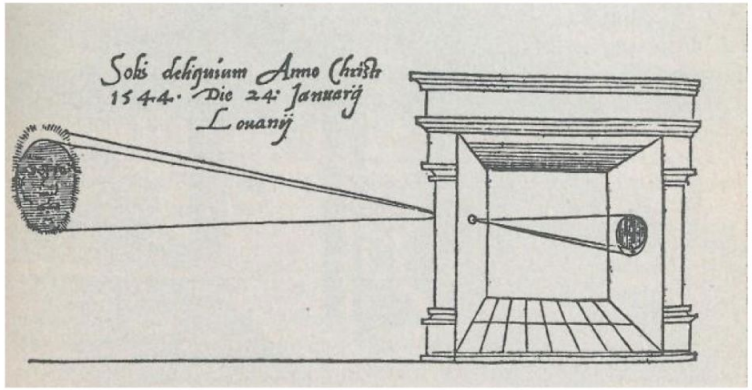
A brief history of computer vision and deep learning



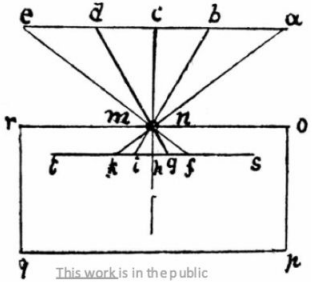
A brief history of computer vision and deep learning

Camera Obscura

Gemma Frisius, 1545



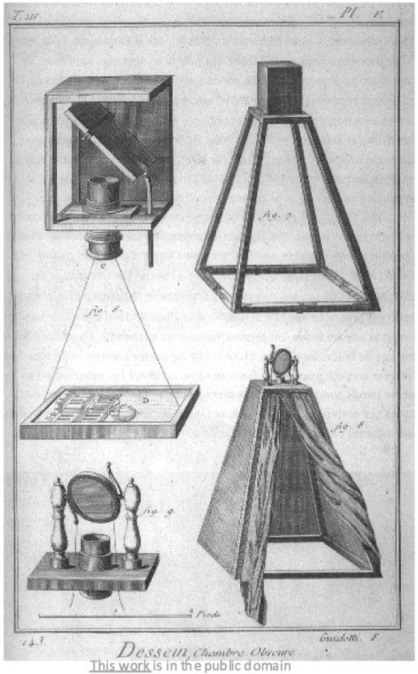
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Leonardo da Vinci,
16th Century AD

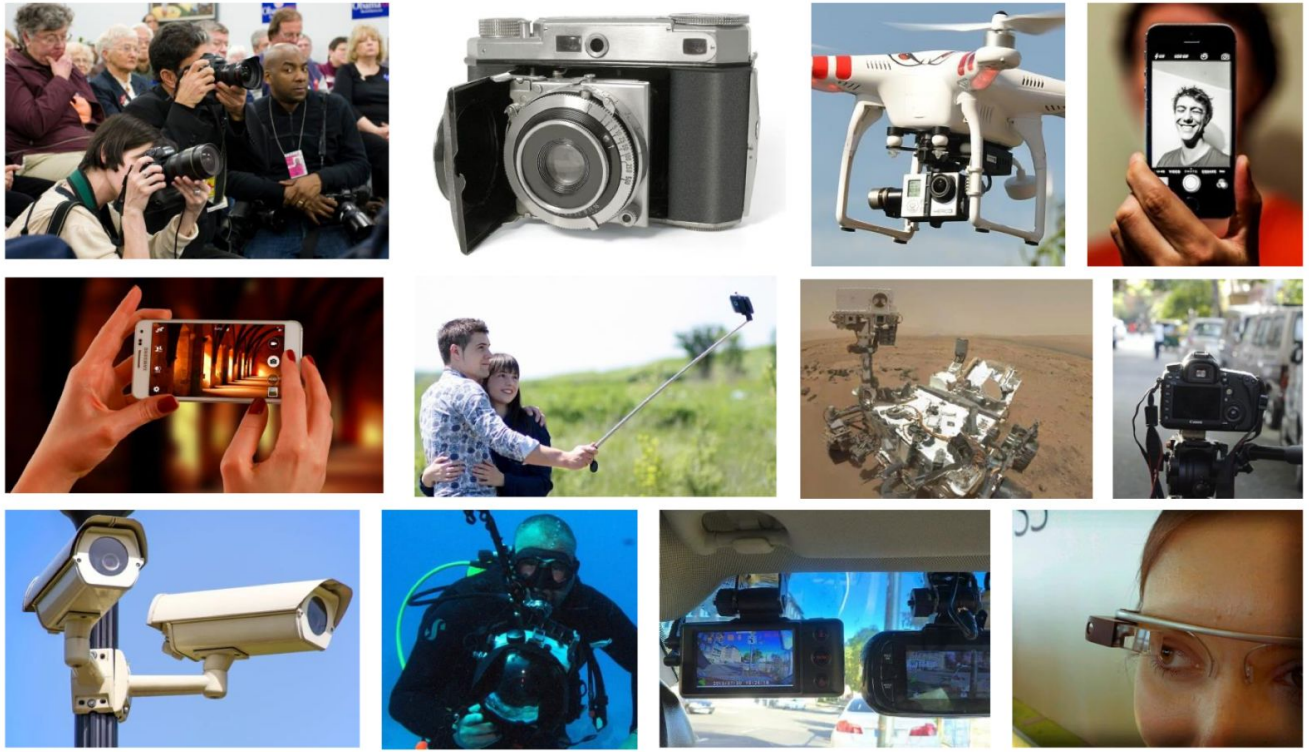
Encyclopedia, 18th Century



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A brief history of computer vision and deep learning

Computer Vision is everywhere!



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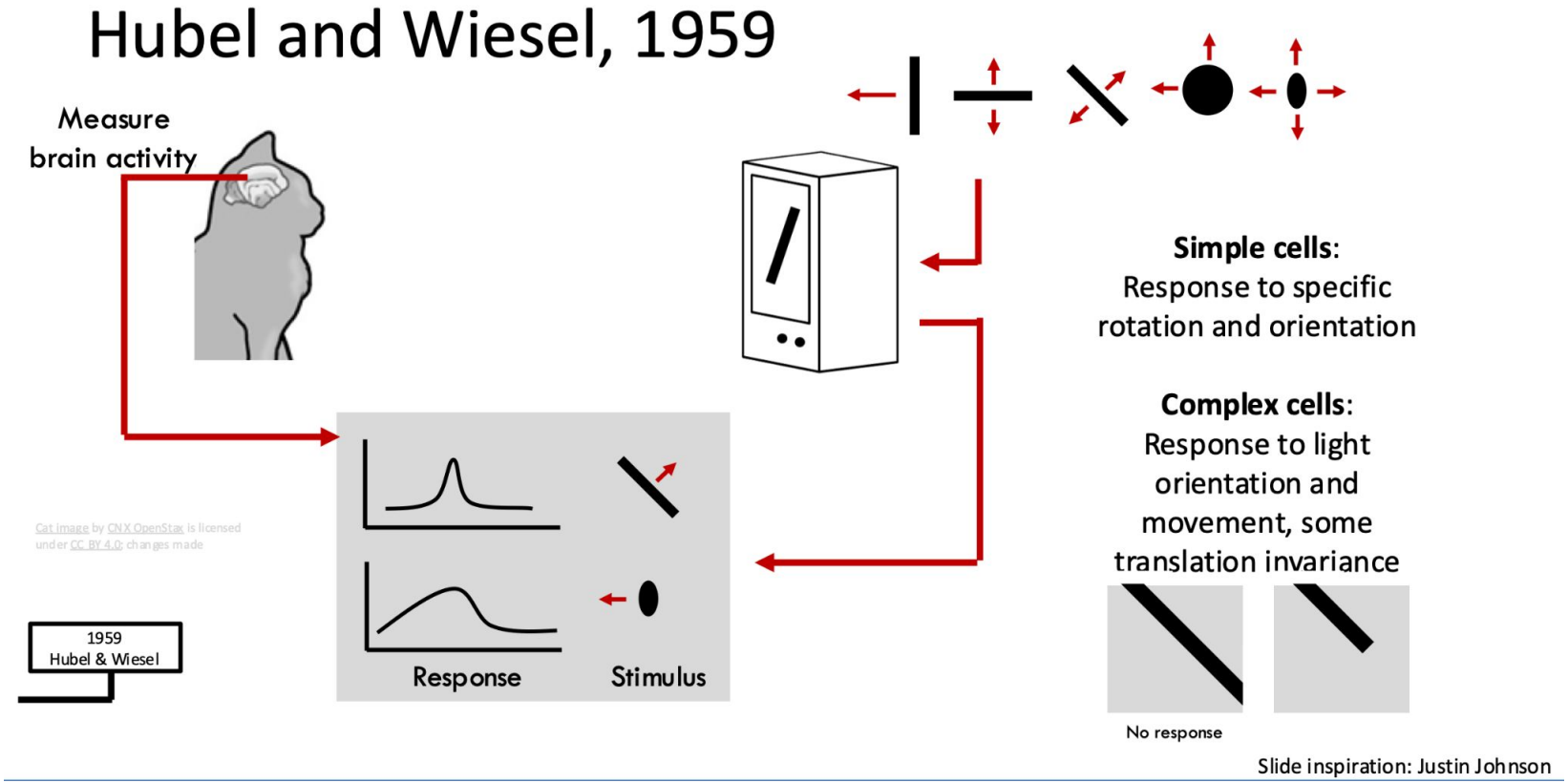
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A brief history of computer vision and deep learning

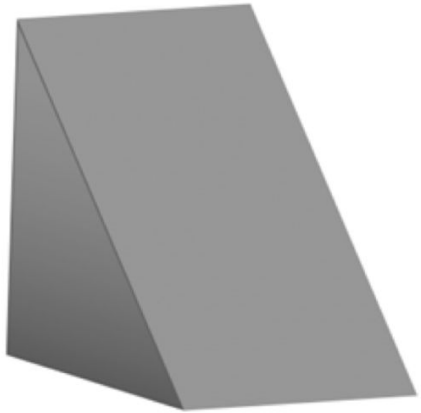
Where did we come from?

A brief history of computer vision and deep learning

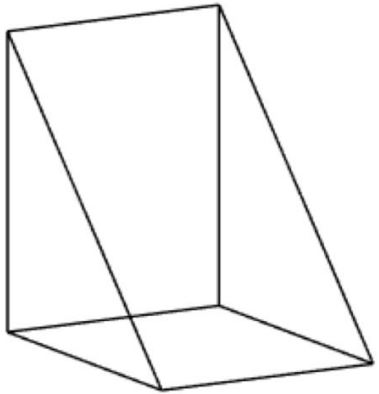


A brief history of computer vision and deep learning

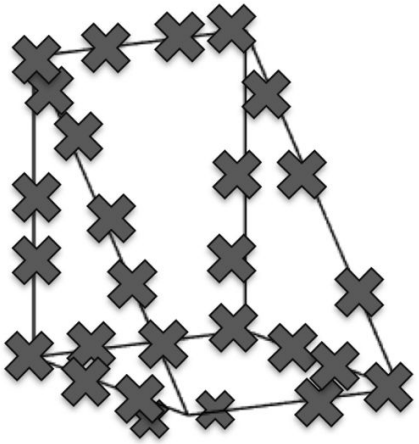
Larry Roberts, 1963



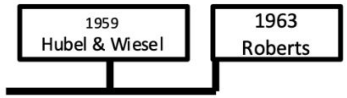
(a) Original picture



(b) Differentiated picture



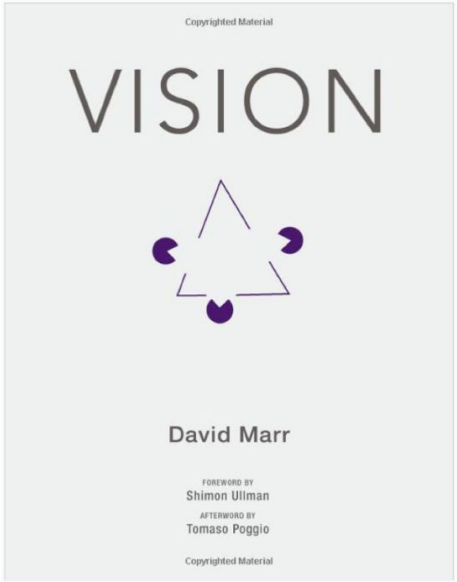
(c) Feature points selected



Lawrence Gilman Roberts, "Machine Perception of Three-Dimensional Solids", 1963

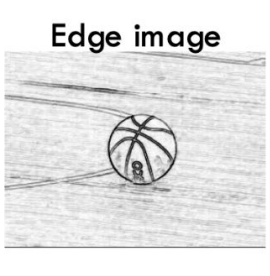
Slide inspiration: Justin Johnson

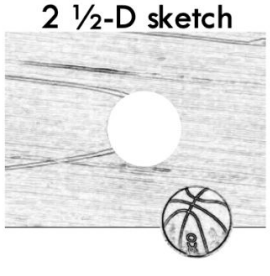
A brief history of computer vision and deep learning





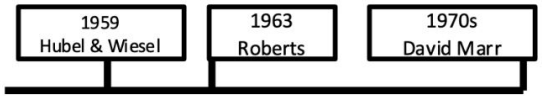
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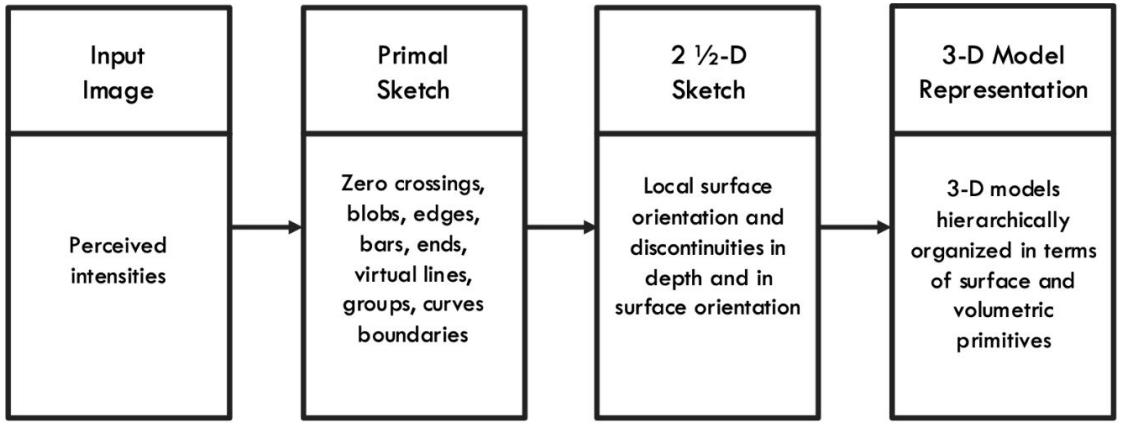






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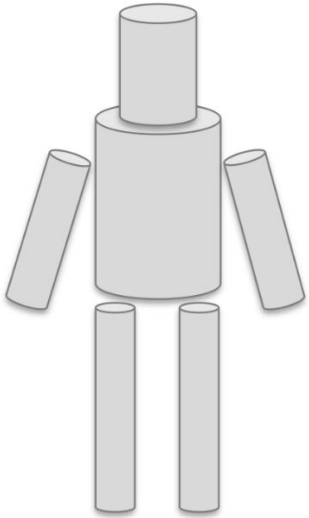


Stages of Visual Representation, David Marr, 1970s

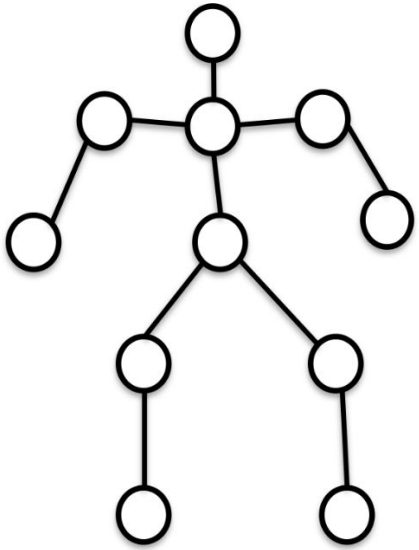
Slide inspiration: Justin Johnson

A brief history of computer vision and deep learning

Recognition via Parts (1970s)



Generalized Cylinders,
Brooks and Binford,
1979



Pictorial Structures,
Fischler and Elshlager, 1973



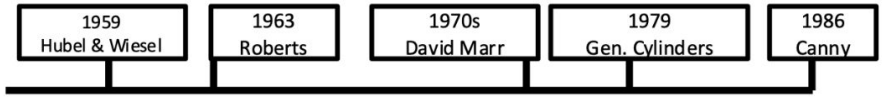
Slide inspiration: Justin Johnson

A brief history of computer vision and deep learning

Recognition via Edge Detection (1980s)







John Canny, 1986
David Lowe, 1987

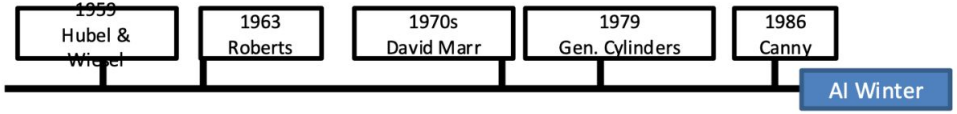
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Slide inspiration: Justin Johnson

A brief history of computer vision and deep learning

Arriving at an “AI winter”

- Enthusiasm (and funding!) for AI research dwindled
- ”Expert Systems” failed to deliver on their promises
- But subfields of AI continues to grow
  - Computer vision, NLP, robotics, compbio, etc.



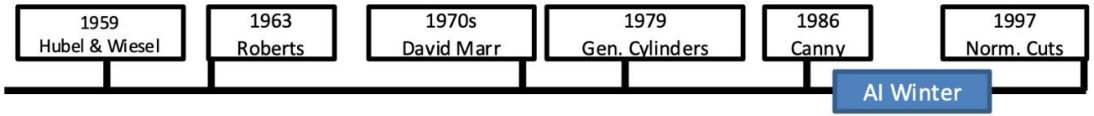
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Slide inspiration: Justin Johnson

A brief history of computer vision and deep learning

Recognition via Grouping (1990s)





Normalized Cuts, Shi and Malik, 1997

Left Image is CC BY 3.0 Middl Image is public Right Image is CC-BY 2.0; changes made

Slide inspiration: Justin Johnson

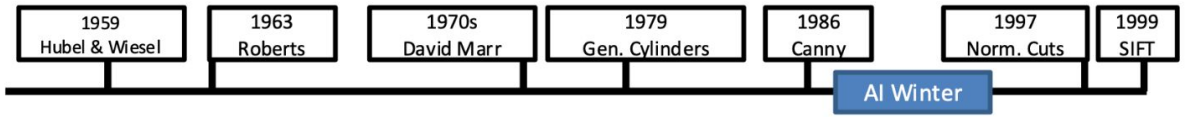
A brief history of computer vision and deep learning

Recognition via Matching (2000s)



Image is public domain

Image is public domain



SIFT, David Lowe, 1999

Slide inspiration: Justin Johnson

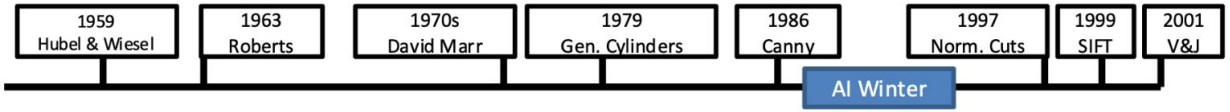
A brief history of computer vision and deep learning

Face Detection

Viola and Jones, 2001

One of the first successful applications of machine learning to vision





Slide inspiration: Justin Johnson

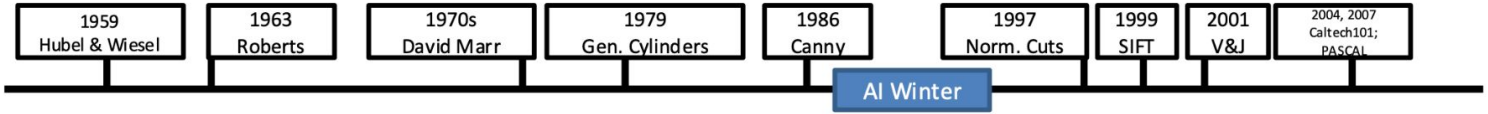
A brief history of computer vision and deep learning

Caltech 101 images



PASCAL Visual Object Challenge





Slide inspiration: Justin Johnson

A brief history of computer vision and deep learning

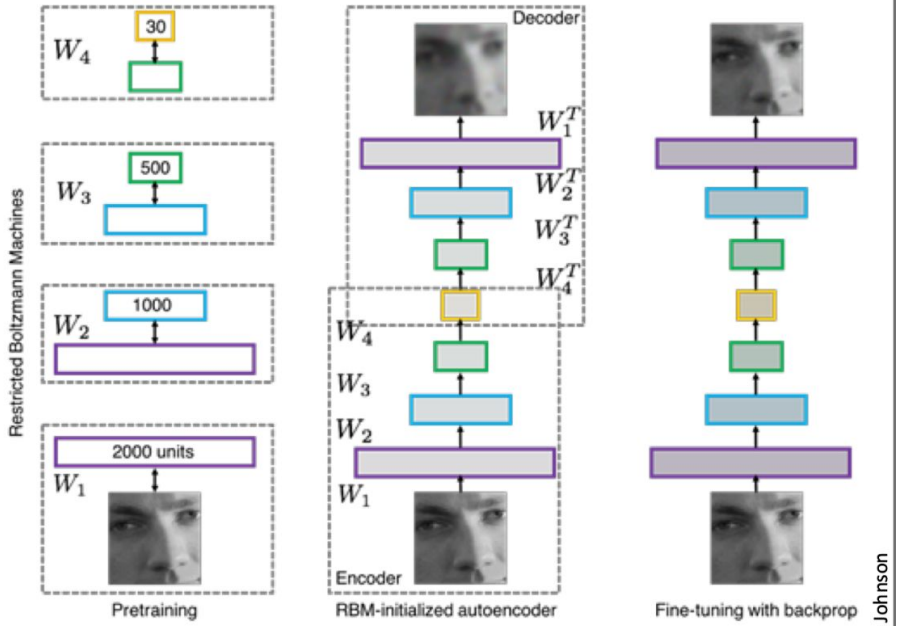
2000s: “Deep Learning”

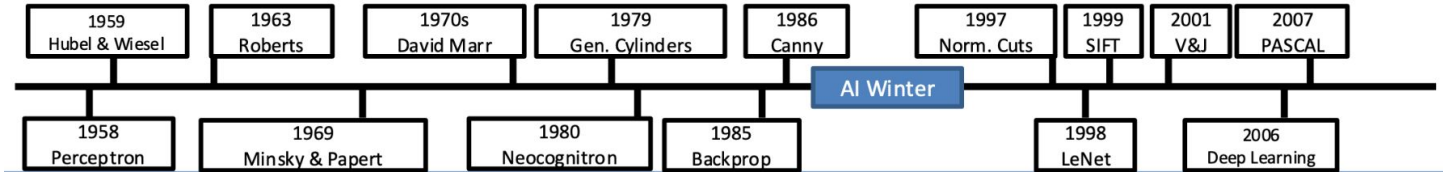
People tried to train neural networks that were deeper and deeper

Not a mainstream research topic at this time

No good dataset to work on

Hinton and Salakhutdinov, 2006
Bengio et al, 2007
Lee et al, 2009
Glorot and Bengio, 2010





Slide inspiration: Justin Johnson

A brief history of computer vision and deep learning

IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:

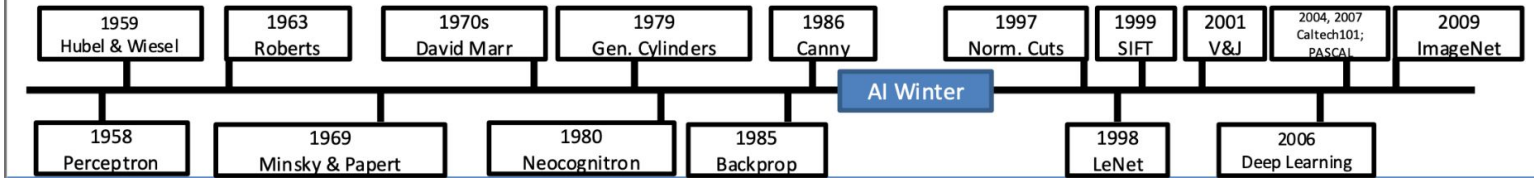
1,000 object classes

1,431,167 images



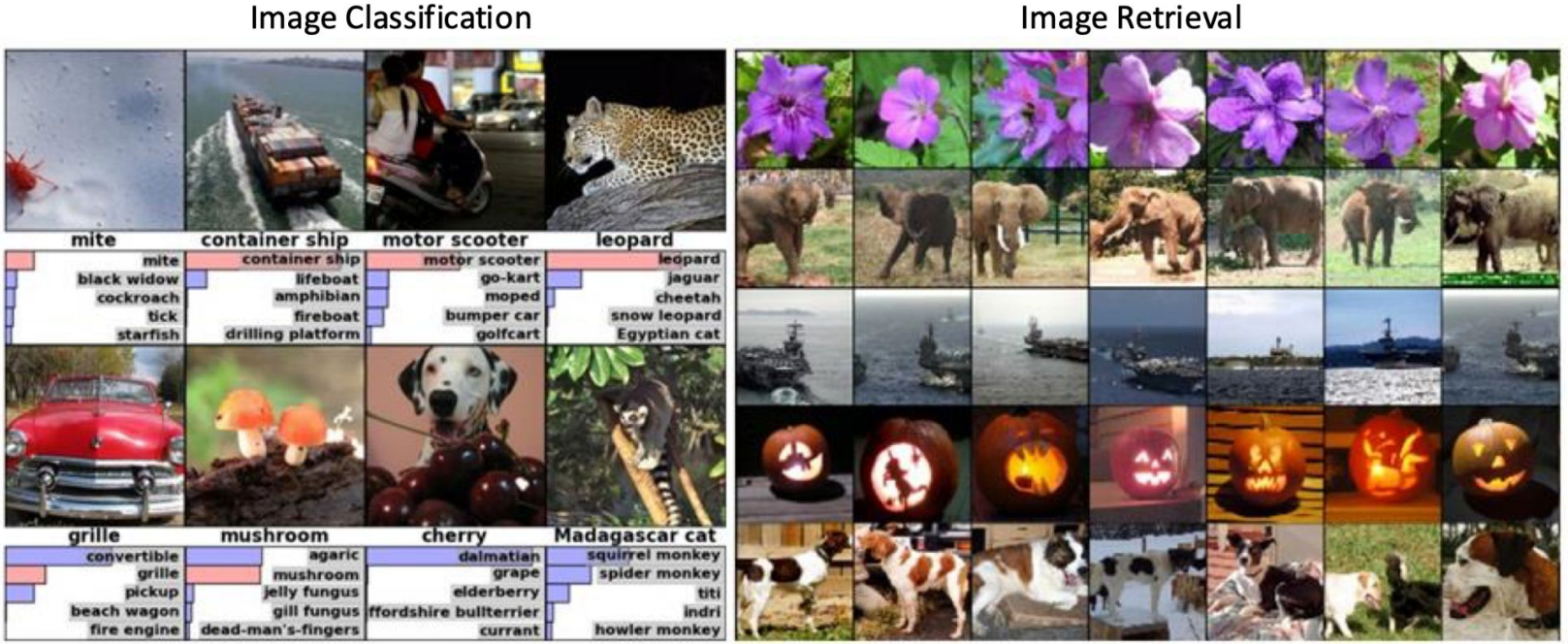
Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle

Deng et al, 2009
Russakovsky et al. IJCV 2015



A brief history of computer vision and deep learning

2012 to Present: Deep Learning is Everywhere



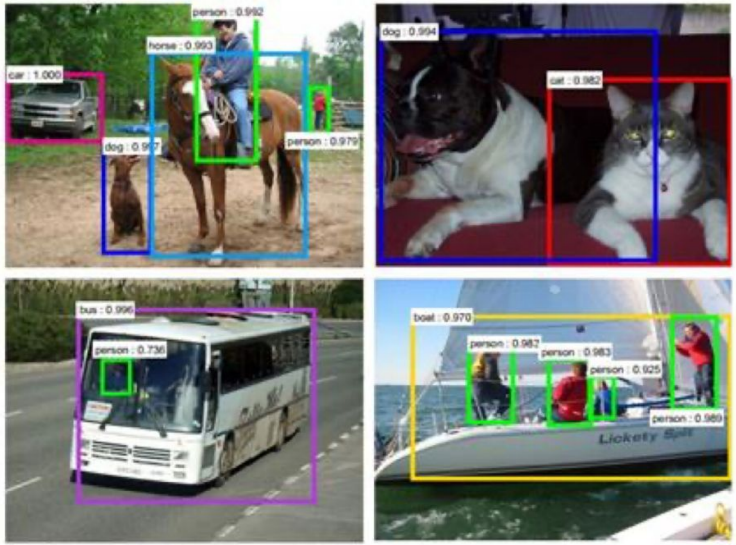
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Slide inspiration: Justin Johnson

A brief history of computer vision and deep learning

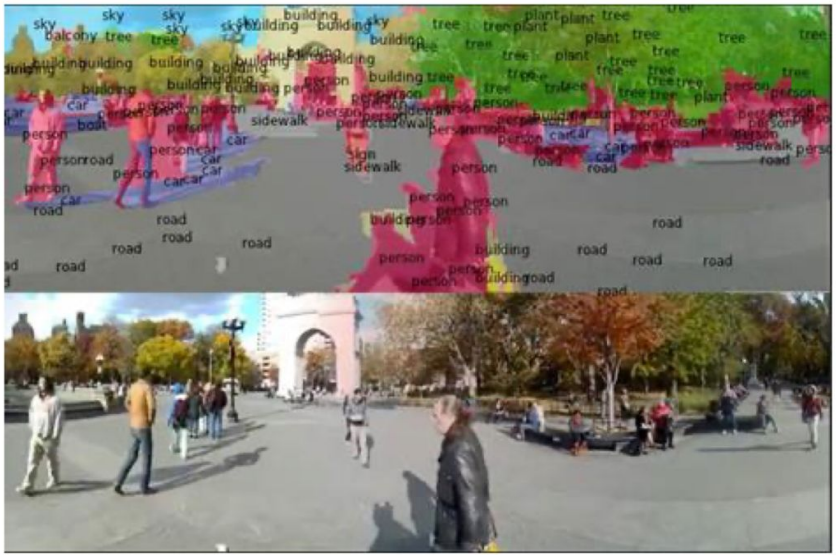
2012 to Present: Deep Learning is Everywhere

Object Detection



Ren, He, Girshick, and Sun, 2015

Image Segmentation



Fabaret et al, 2012

Slide inspiration: Justin Johnson

A brief history of computer vision and deep learning

2012 to Present: Deep Learning is Everywhere

Year 2010

NEC-UIUC



Dense descriptor grid: HOG, LBP

Coding: local coordinate, super-vector

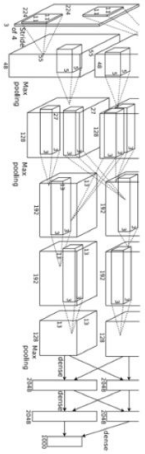
Pooling, SPM

Linear SVM

[Lin CVPR 2011]

Year 2012

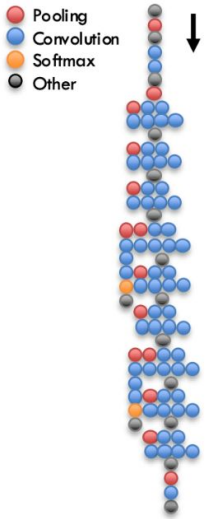
SuperVision



[Krizhevsky NIPS 2012]

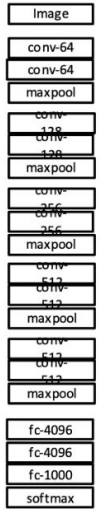
Year 2014

GoogLeNet



[Szegedy arxiv 2014]

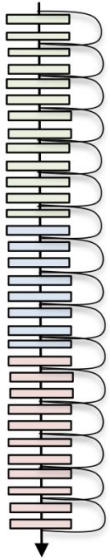
VGG



[Simonyan arxiv 2014]

Year 2015

MSRA



[He ICCV 2015]

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Today’s agenda

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Visual perception and understanding: What makes a good visual representations?

Perceiving and Understanding the Visual World



- Informative and compact?
- Generalizable and robust?
- Unbiased?
- Good performance in downstream applications?
- Close to human perception?
-

These are all true, but without a definite answer…

Today’s agenda

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Classic image processing pipeline: from pixels to features

Raw images:

- Preprocessing (e.g., grayscale, resize, normalize)
- Handcrafted feature extraction (e.g., edges, corners, texture descriptors)
- Shallow classifier (e.g., SVM, Logistic Regression)

Examples of handcrafted features:

- **Edge detectors** – e.g., Sobel, Canny
- **Texture descriptors** – e.g., Gabor filters, LBP
- **Shape features** – e.g., HOG (Histogram of Oriented Gradients)

Case study: HOG (Histogram of Gradients, 2005)

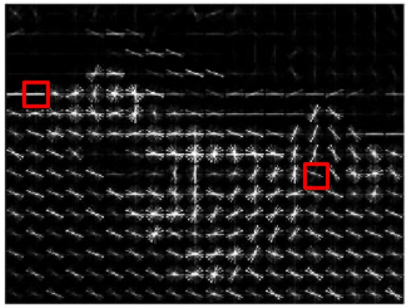
- Shape Based HOG
  - Uses magnitude and direction of the gradient to generate histograms of gradients in a region of an image
  - Captures shapes and contours well, robust to small changes in appearance, and works well for structured patterns like humans, animals, or traffic signs

📌 **Key Steps:**

1. **Divide** the image into small spatial regions (*cells*)
2. **Compute gradients** (direction + magnitude) for each pixel
3. **Histogram** of gradient directions in each cell
4. **Normalize** over larger blocks for better illumination invariance



Divide image into 8x8 pixel regions
Within each region quantize edge direction into 9 bins



Example: 320x240 image gets divided into 40x30 bins; in each bin there are 9 numbers so feature vector has 30*40*9 = 10,800 numbers

Lowe, "Object recognition from local scale-invariant features", ICCV 1999
Dalal and Triggs, "Histograms of oriented gradients for human detection," CVPR 2005

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Deep Learning Basics

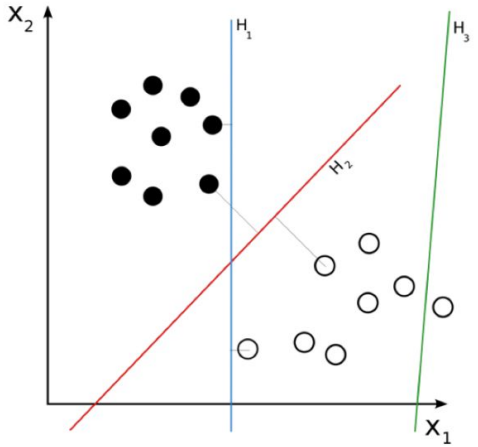
- Image Classification: A core task in Computer Vision



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Linear Classifier

Deep Learning Basics

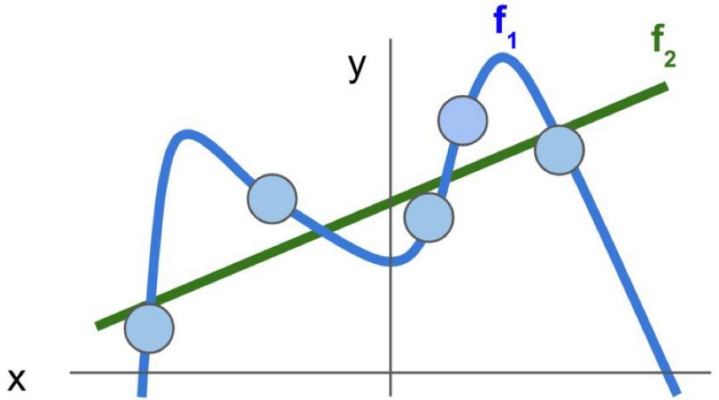
- Image Classification: A core task in Computer Vision



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Regularization & Optimization

Convolutional Neural Networks

- Introduced by Yann LeCun and Yoshua Bengio (1995)
- Neuro-biologically motivated by the findings of locally sensitive and orientation-selective nerve cells in the visual cortex.
- They designed a network structure that implicitly extracts relevant features.
- Convolutional Neural Networks are a special kind of multi-layer neural networks.

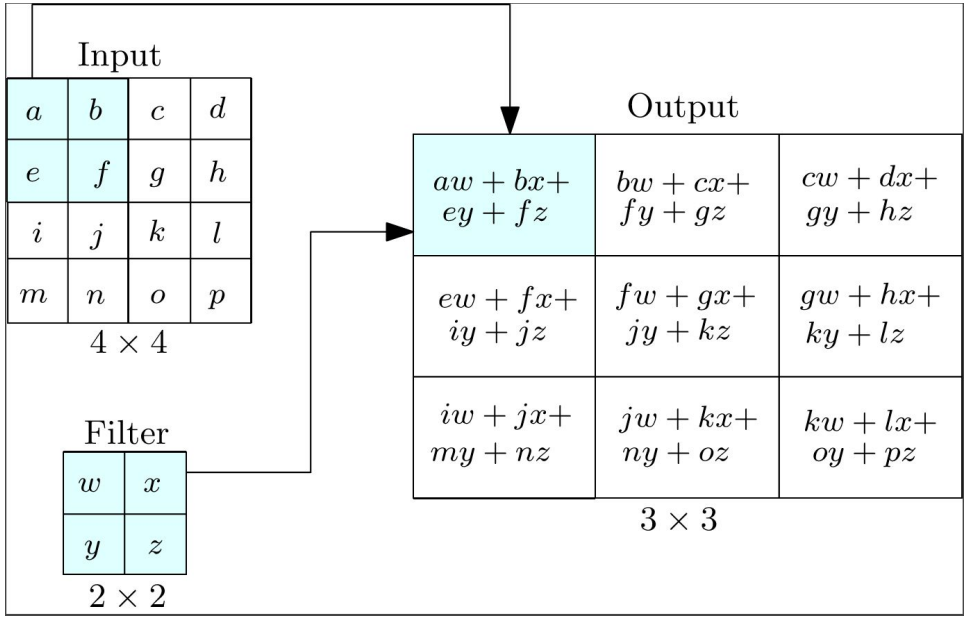
Paper ref: LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. (1998). Gradient-based learning applied to document recognition. Proceedings of the IEEE, 86(11) 2278-2324. 3, 4

Convolutional Neural Networks

- Inspired by studies of the cat's visual cortex [1], developed in computer vision to work on regular grids such as images [2].
- Feed-forward NNs , each neuron receives input from a neighborhood of the neurons (receptive fields) in the previous layer.
- Receptive fields, allow CNNs to recognize complex patterns in a hierarchical way, by combining lower-level, elementary features into higher-level features *compositionality*.
  - raw pixels=> edges =>shapes =>objects.
- Absolute positions of features in the image are not important – only useful respective positions is useful composing higher-level patterns.
- Model detect a feature regardless of its position in the image - local invariance.
- Compositionality, local invariance two key concepts of CNNs.

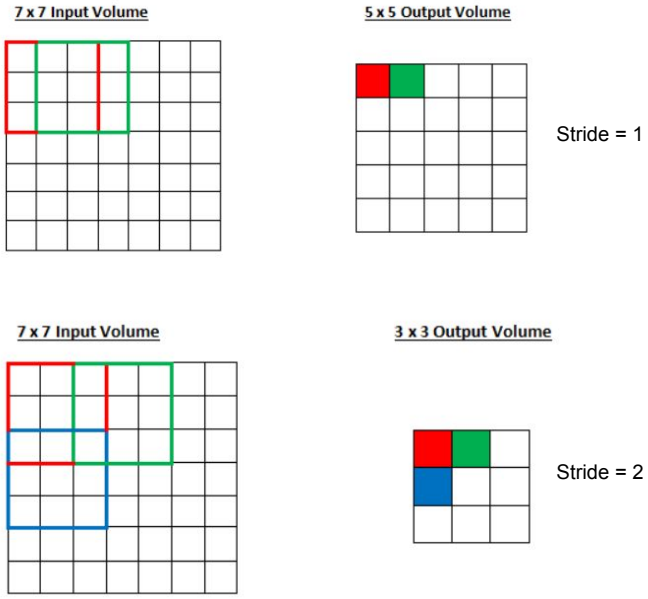
[1] Hubel, David H., and Torsten N. Wiesel (1962). Receptive fields, binocular interaction and functional architecture in the cat's visual cortex. The Journal of physiology 160.1:106-154. 4
[2] LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. (1998). Gradient-based learning applied to document recognition. Proceedings of the IEEE, 86(11), 2278-2324. 3, 4

Convolutional Neural Networks



- Convolution exploits spatial local correlations in the feature space by enforcing local connectivity pattern between neurons of adjacent layers
- Drastic reduction of free parameters compared to fully connected network reducing overfitting and computational complexity of the network

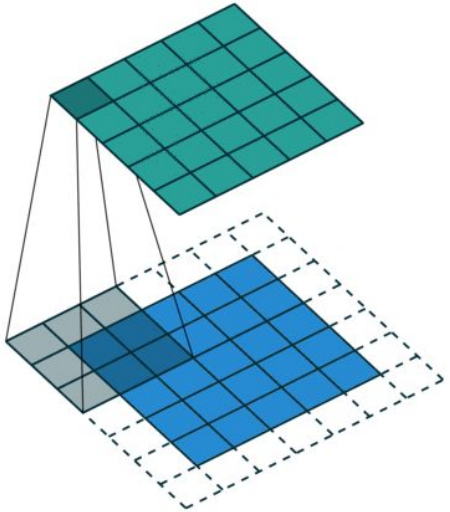
Convolutional Neural Networks



- **Kernel/Filter Size**: defines the field of view of the convolution. A common choice for 2D is 3x3 pixels.

- **Stride**: step size of the kernel when traversing the image. default : 1, we stride=2 downsampling an image - similar to MaxPooling.

- **Padding**: defines how the border of a sample is handled.



CNN architectures

Convolution and **pooling** operators extract features while respecting 2D image structure

Fully-Connected layers form an MLP at the end to predict scores

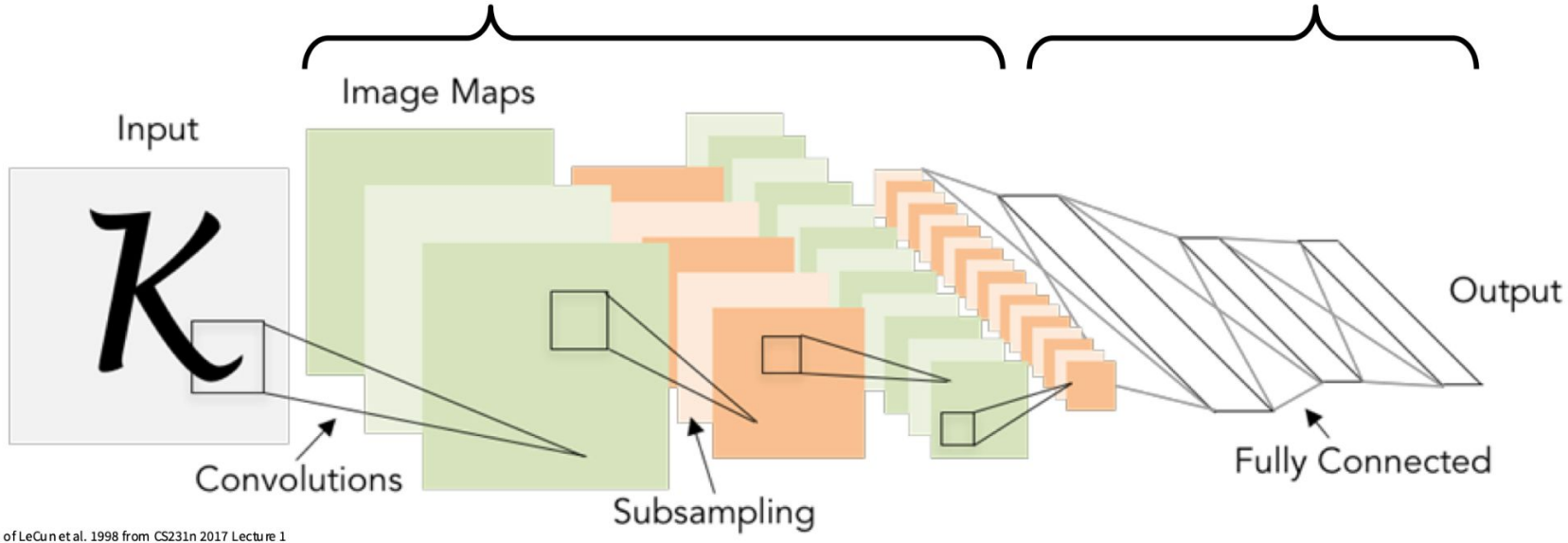


Illustration of LeCun et al. 1998 from CS231n 2017 Lecture 1

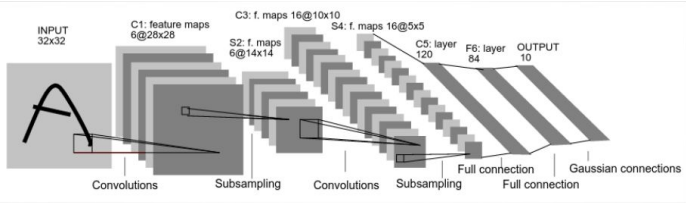
Feature maps

- Feature Map - Obtained by convolution of the feature matrix with a linear filter, adding a bias term and applying a non-linear function
- Non-linear functions:
  - Sigmoid $\frac{1}{1+e^{-x}}$
  - Tanh $\frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}$
  - Rectified Linear Unit (ReLU) -> Most popular choice avoids saturation issues, makes learning faster $f(x)=\max(0,x)$

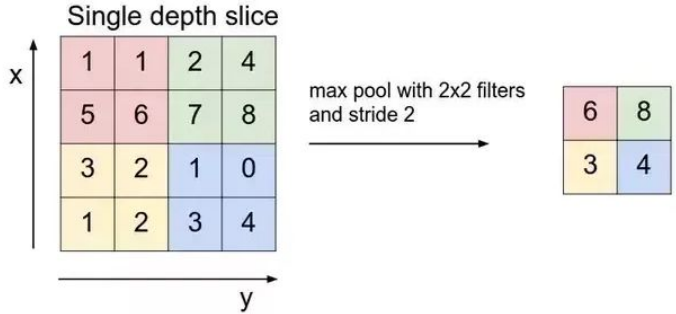
- Require a number of such feature maps at each layer to capture sufficient features

Pooling

- Sub-sampling layer
- Variants:
  - Max pooling
  - Weighted average
  - L2 norm of neighborhood
- Provides translation invariance
- Reduces computation



Lenet-5 (Lecun-98), Convolutional Neural Network for digits recognition



A bit of history:

ImageNet Classification with Deep Convolutional Neural Networks [Krizhevsky, Sutskever, Hinton, 2012]



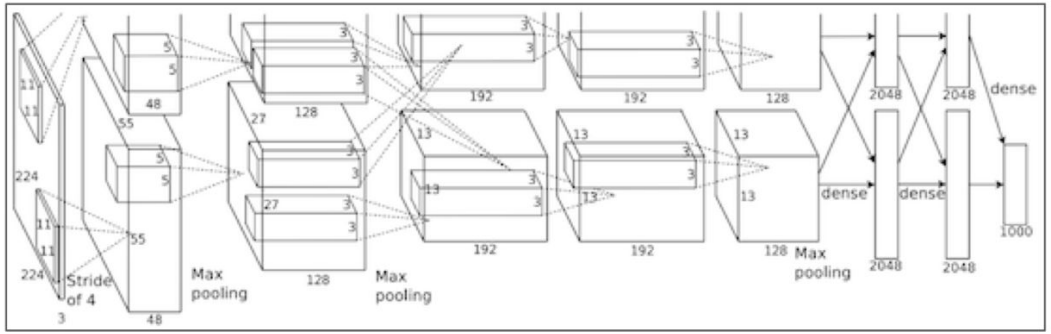
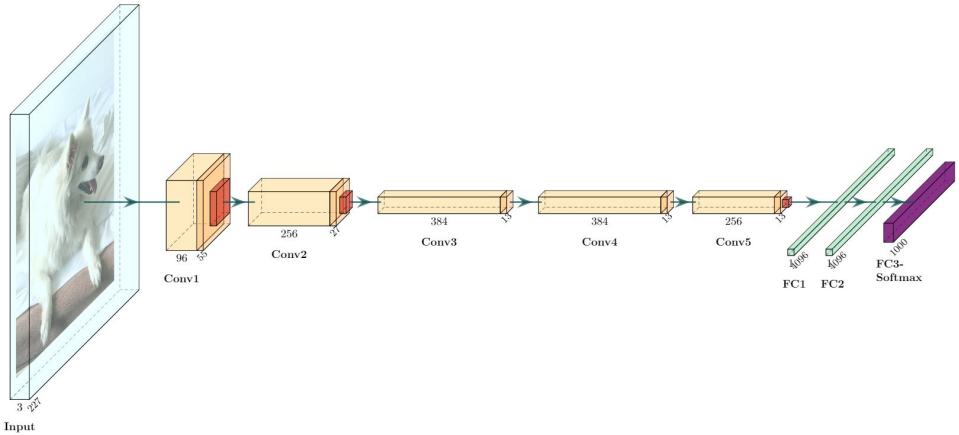


Figure copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

"AlexNet"

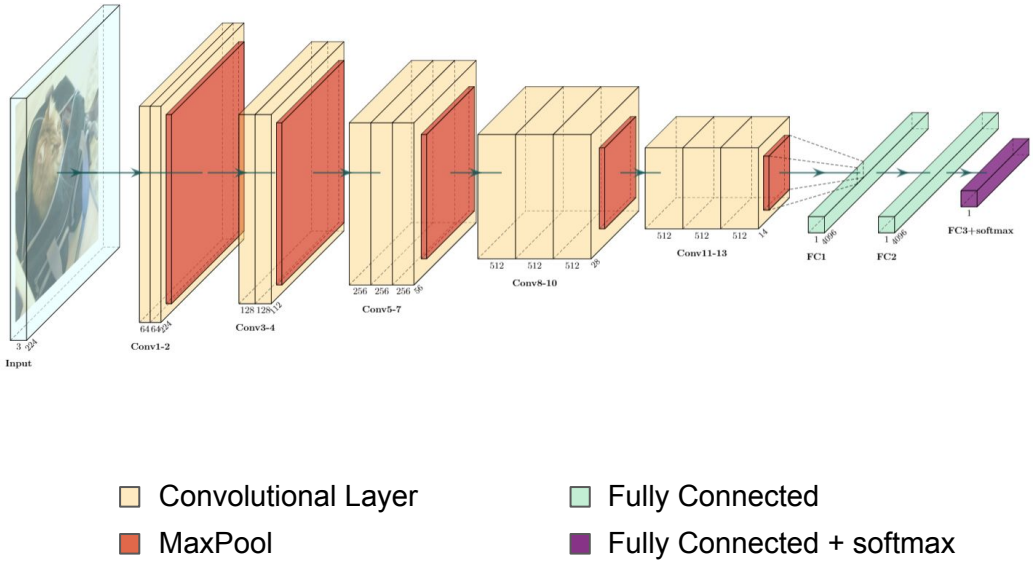
AlexNet

- AlexNet
  - 5 convolutional layers
    - 96 filters, kernel size (11,11), stride (4,4)
    - 256 filters, kernel size (5,5), stride (2,2)
    - 384 filters, kernel size (3,3), stride(1,1)
    - 384 filters, kernel size (3,3), stride(1,1)
    - 256 filters, kernel size (3,3), stride(1,1)
  - 3 Fully Connected Layers
    - 2x4096 nodes
    - Softmax with nodes = # of classes



VGG series

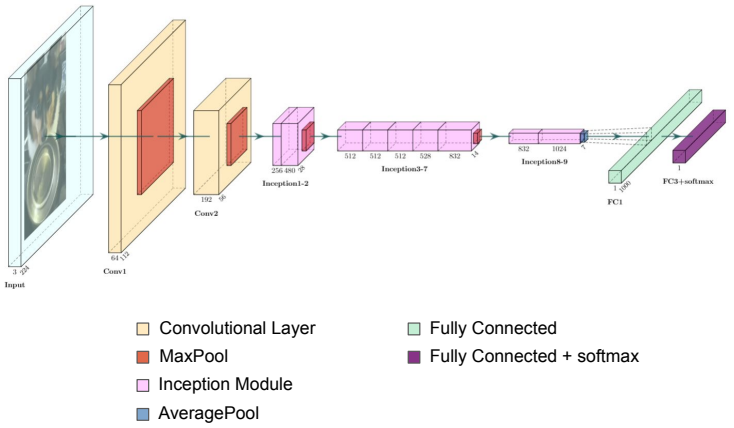
- **VGG16** by Simonyan and Zisserman
  - Competed in ILSVRC 2014 and achieved a **top-5 error of 7.3%**
  - Fixed kernel size to (3,3) and stride to (1,1). Pooling is fixed to size (2,2) with stride (2,2).
  - Despite the large amount of parameters (138 million), the simple architecture and small amount of hyperparameters makes VGG16 easy to use
  - remains popular to this day as a base line



Karen Simonyan and Andrew Zisserman. "Very Deep Convolutional Networks for Large-Scale Image Recognition". In International Conference on Learning Representations. 2015

GoogLeNet

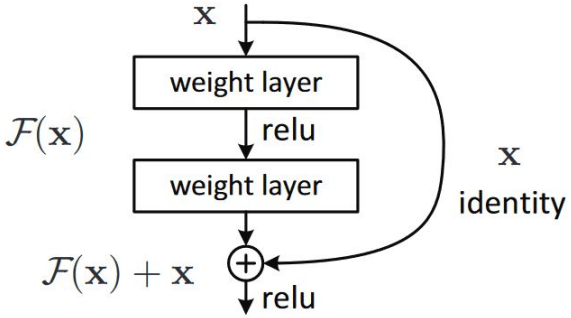
- **GoogLeNet** (a.k.a Inception-v1) by Szegedy et al.
  - Also competed in ILSVRC 2014 and achieved a **top-5 error of 6.67%**
  - Introduces the Inception module, repeating blocks of convolutional layers with 3 branches that are combined at the end of the block.
  - Uses a 1x1 identity convolution for dimensionality reduction
    - when used before a convolutional layer with a larger kernel, it uses less convolutions than directly reducing dimensions
  - *Fast and lightweight model, 5 million parameters*



References: Christian Szegedy et al. “Going deeper with convolutions”. In:2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). 2015

ResNet

- As image networks become deeper, we run into the problem of vanishing gradients
  - Gradients become too small during backpropagation, earlier layers might not train
- ResNet by He et al. introduce residual connections to solve this
  - Allow information (and gradients) to skip over some convolutional layers
    - These skips make the network train as if it is smaller
    - Sometimes called skip connections!
  - Competed in ILSVRC 2015 and achieved a **top-5 error of 3.57%**



Kaiming He et al. "Deep Residual Learning for Image Recognition". In:2016 IEEE Conference on Computer Vision and Pattern Recognition(CVPR). 2016

Image classification - Benchmarks

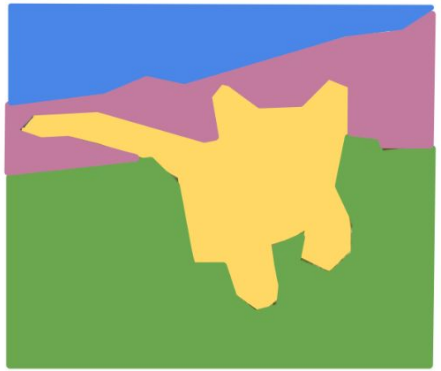
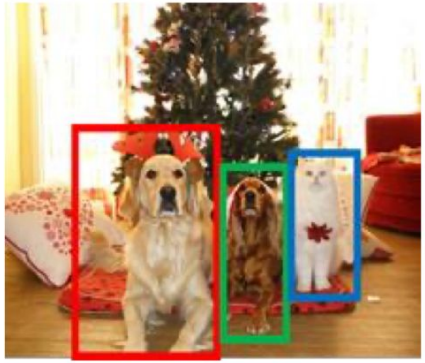
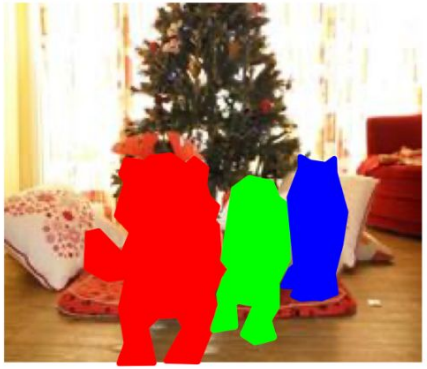
- MNIST
  - 10 classes of handwritten digits
  - 60k training images, 10k test images
  - Good starting dataset, simple subjects, small images, greyscale
- CIFAR-10
  - 10 object classes
  - 50k training images, 10k test images
  - More complicated subjects, small images, full color
- ImageNet
  - 1000 object classes
    - (1) *image-level annotation* of a binary label - presence or absence of an object class, e.g., “there are cars in this image” but “there are no tigers,”
    - (2) *object-level annotation* - a tight bounding box and class label around an object instance in the image, e.g., *“there is a screwdriver centered at position (20,25) with width of 50 pixels and height of 30 pixels*”.
  - 1,281,167 training images, 50k validation images, 100k test images.
  - Most “serious” dataset, varied image sizes, full color

Today’s agenda

- A brief history of computer vision and deep learning
- Visual perception and understanding: What makes a good visual representation? *(Lab to follow after the lecture!)*
- Classic image processing techniques (a quick touch and recap)
- Image classification and beyond
- Object detection and segmentation
- Other more advanced topics on visual and multimodal generation

Object detection and segmentation

Tasks Beyond Image Classification

Classification	Semantic Segmentation	Object Detection	Instance Segmentation
			
CAT	GRASS, CAT, TREE, SKY	DOG, DOG, CAT	DOG, DOG, CAT
No spatial extent	No objects, just pixels	Multiple Object	

This image is CC0 public domain

Classic model architectures

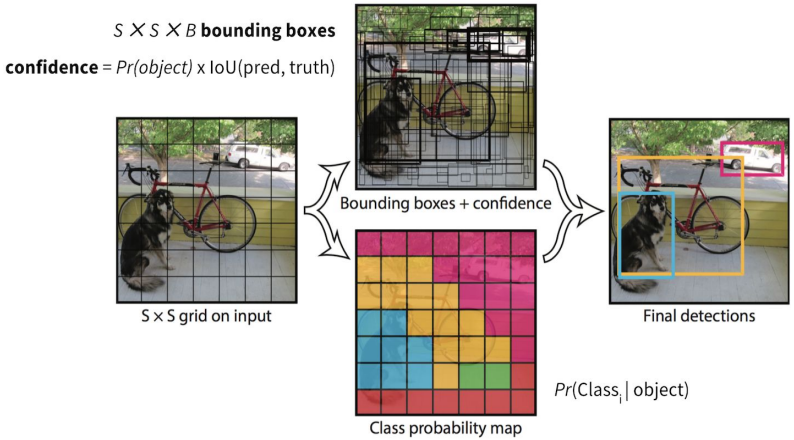
- R-CNN
- Fast R-CNN, Faster R-CNN
- YOLO
- Benchmarks

Object detection

- To account for bounding boxes, mean average precision (mAP) is used as the standard metric
  - Average Precision (AP)
    - Precision measures how many of your positive predictions are correct TP/(TP+FP)
    - Recall measures how well you find all the positives TP/(TP+FN)
    - Average Precision is the area under the precision-recall curve
  - Intersection over Union (IoU)
    - Determines how much the predicted bounding box has to overlap with the ground truth box in order to count as correct
    - The same predicted box can be correct or incorrect depending on the IoU threshold
  - mAP
    - Typically competition or benchmark defines a set of acceptable IoU thresholds (ex [0.5, 0.6, 0.7, 0.8, 0.9])
    - Find the AP for each IoU and take the mean

Object detection: two-stage approaches

- Two-stage object detectors
  - Pipeline: *proposal stage* ⇔ classification stage
    - *Proposal stage*
      - Examines the entire image and identifies *Regions of Interest (RoIs)*, areas that *might* contain an object
      - Each region can be considered a bounding box
    - *Classification stage* - looks at these RoIs and classifies them



References:
https://lilianweng.github.io/lil-log/2018/12/27/object-detection-part-4.html

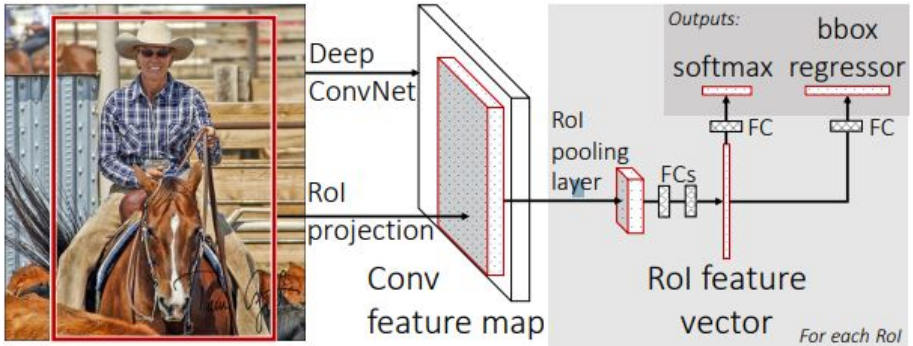
Object detection: two-stage approaches

- R-CNN by Girshick et al.
  - *The first to popularize the two stage approach for object detection*
  - Achieves ***53.7% mAP on PASCAL-VOC2010***, an 18% improvement over the previous best
  - Achieves ***31.4% mAP on ISLVRC2013*** object detection dataset, a 7.1% improvement
  - Proposal stage
    - Use *selective search algorithms* to identify 2,000 RoIs per image
  - Classification stage
    - AlexNet for feature extraction of each RoIs
    - Features fed into many binary support vector machines (SVMs) for classification
      - 1 SVM per class (i.e. cat, bird)
  - Extra bounding box tuning stage using a regression model
- Speed bottleneck
  - Three models are very expensive and slow to train
  - Can take up to 49s per inference!

References:
Ross Girshick et al. "Rich feature hierarchies for accurate object detection and semantic segmentation". In:2014 IEEE Conference on Computer Vision and Pattern Recognition. 2014.

Object detection: two-stage approaches

- Fast-RCNN by Girshick
  - Achieves **mAP 68.8% on PASCAL-VOC2010** and **speeds up inference time to 2.3s**
  - Combined feature matrix
    - Extract features over the entire image with a CNN (AlexNet, VGG16, etc) first
    - Features are used for both the *object classifier* and *bounding-box regression*
  - RoI Pooling
    - Generate RoIs over the feature matrix using selective search
    - Split each RoI into a grid and do max pooling on each grid space to generate a feature vector
  - Two outputs
    - Softmax classification for each ROI
    - Bounding box regression model to predict the offset of the ROIs when scaled up from the feature vector
- Fast! But selective search is still a bottleneck

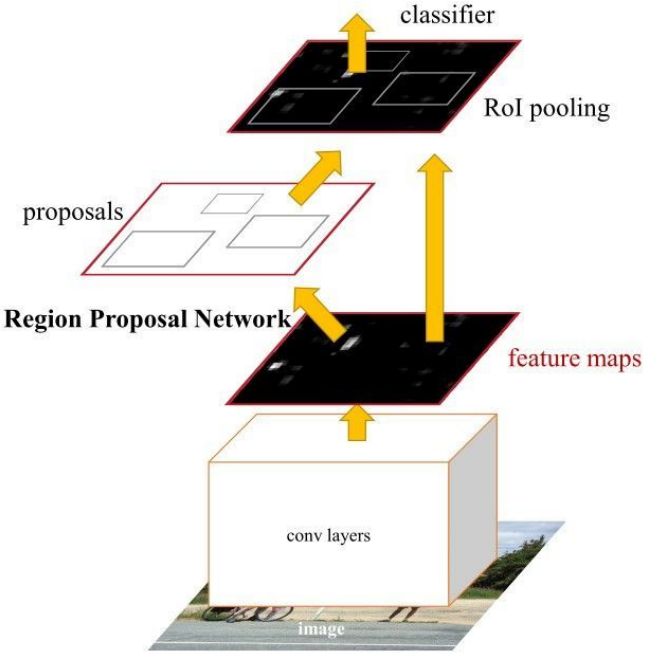


References:

Ross Girshick. "Fast r-cnn". In:2015 IEEE International Conference on Computer Vision (ICCV). 2015

Object detection: two-stage approaches

- Faster-RCNN by Ren et al.
  - Achieves an **mAP of 75.9% on PASCAL-VOC2012** and **inference time of 200ms**
  - Replace selective search with a region proposal network (RPN) that predicts RoIs directly over the feature matrix generated by a CNN.
  - RPN and classification models can share layers



References:
Shaoqing Ren et al. "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks". In:IEEE Transactions on Pattern Analysis and Machine Intelligence (2017)

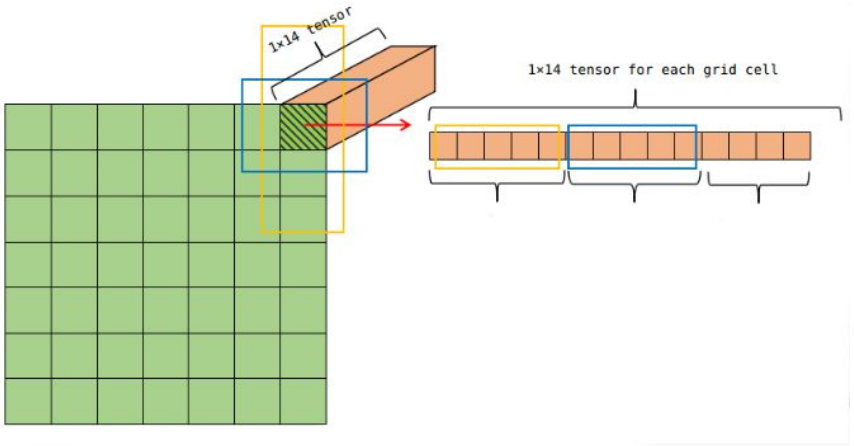
Object detection: one-stage approaches

- *No region proposal stage*
- Directly predict bounding boxes *and* classes over an image
  - Until recently, one stage approaches typically had *lower prediction accuracies* but much *faster inference times* than two stage approaches.
  - With current advancements this is no longer the case!
- You Only Look Once (YOLO) by Redmon et al.
  - First one stage approach
  - Achieves an **mAP of 57.9% on PASCAL-VOC2012**, but only needs **22ms per inference** - as opposed to state of the art *Faster-RCNN - mAP of 75.9%, 200m ms/inference*

References:
Joseph Redmon et al. "You Only Look Once: Unified, Real-Time Object Detection". In:2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). 2016

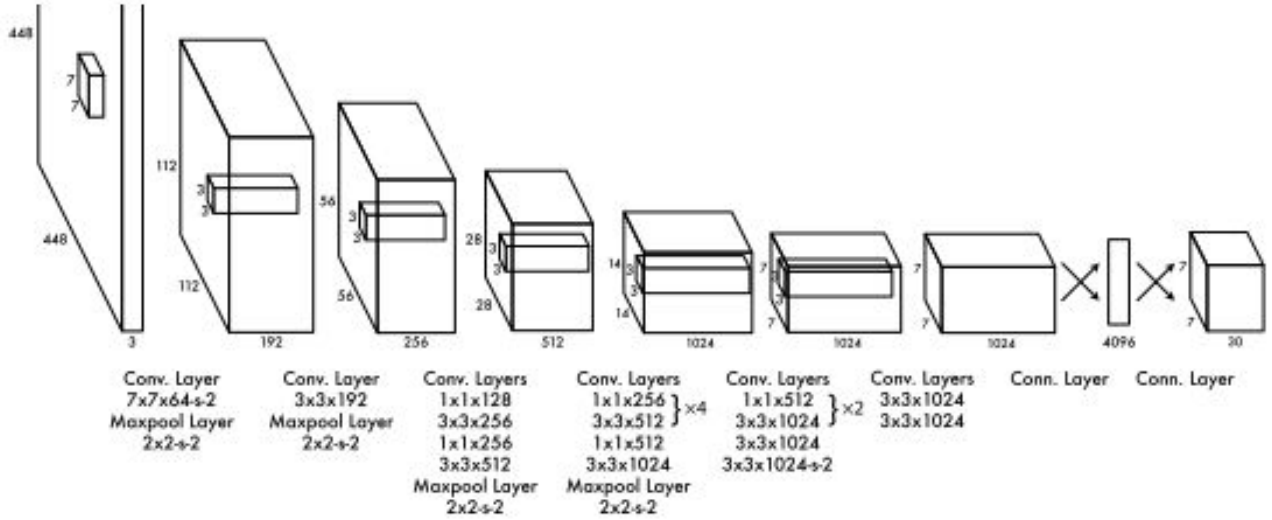
Object detection: one-stage approaches

- YOLO
  - A single CNN predicts the bounding boxes and classes
    - CNN encodes an image into an SxS grid where each cell corresponds to an area of the image
    - If the center of an object is in that area, the corresponding grid cell is supposed to predict it
      - Each cell predicts *B* number of bounding boxes, probability *p* of an object being contained in that box, and a single class for all boxes



Reference: https://zhanghanduo.github.io/post/yolo1/

Yolo architecture



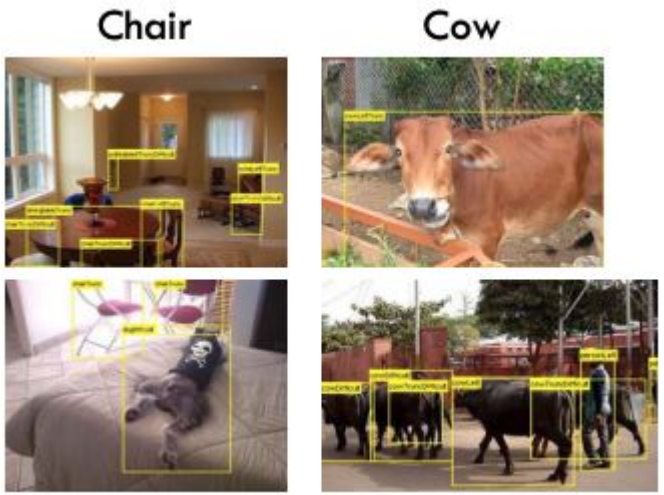
Object detection: one-stage approaches

- A common problem is many overlapping possible bounding boxes
- Use *non-max suppression* (NMS) to filter boxes
  - Select highest confidence bounding box
  - Compute its IoU with other boxes and remove ones that exceed a threshold
  - Repeat until there are no more overlapping boxes



Object Detection - Benchmarks

- Pascal-VOC
  - 20 object classes - classification and detection
  - New datasets released yearly from 2005-2012
    - Number and complexity of annotations increase year to year
  - Most commonly used versions
    - 2010: 10.103 images and 23.374 object instances
    - 2011/2012: 11.530 images and 27.450 object instances
    - 2007: 9.963 images and 24.640 object instances
      - Last year to include test labels!
- Common Objects in Context (COCO)
  - 80 object classes
  - >200K labeled images
  - 1.5 million object instances



Real-Time Detectors	Train	mAP	FPS
100Hz DPM [31]	2007	16.0	100
30Hz DPM [31]	2007	26.1	30
Fast YOLO	2007+2012	52.7	**155**
YOLO	2007+2012	**63.4**	45
Less Than Real-Time			
Fastest DPM [38]	2007	30.4	15
R-CNN Minus R [20]	2007	53.5	6
Fast R-CNN [14]	2007+2012	70.0	0.5
Faster R-CNN VGG-16[28]	2007+2012	73.2	7
Faster R-CNN ZF [28]	2007+2012	62.1	18
YOLO VGG-16	2007+2012	66.4	21

References:

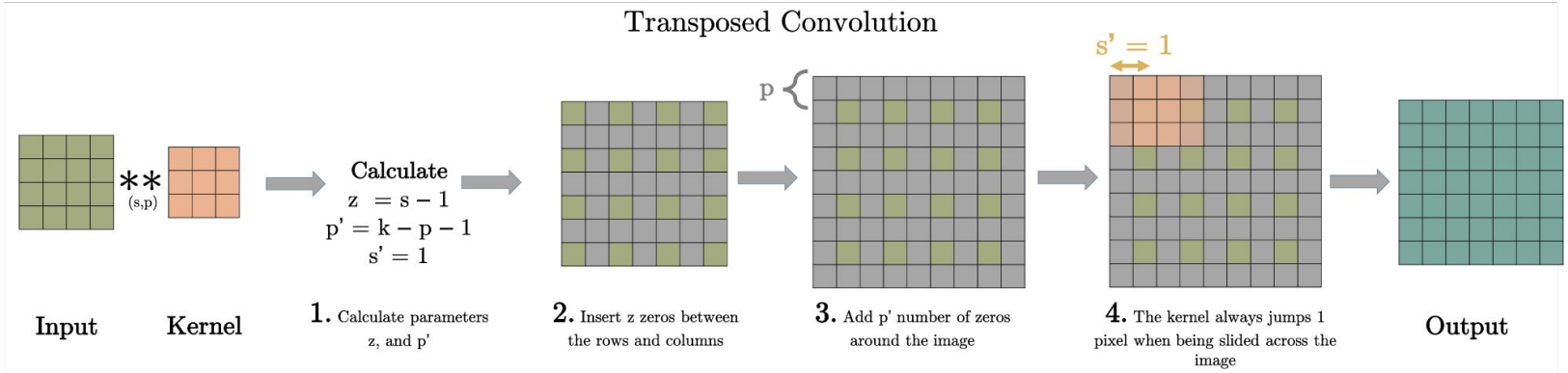
http://host.robots.ox.ac.uk/pascal/VOC/ https://cocodataset.org/#home

Semantic Segmentation - Fully Convolutional Networks

- Since prediction is now pixel-by-pixel, features obtained by repeatedly downsampling are no longer acceptable!
  - Semantic segmentation premise: *keep the output the same resolution as the input*
    - Classification doesn't care about location, object detection can scale the bounding boxes
  - Idea: Use convolutional layers with a stride of 1 to preserve resolution
    - Very, very, computationally expensive
    - Not feasible

Semantic Segmentation - Fully Convolutional Networks

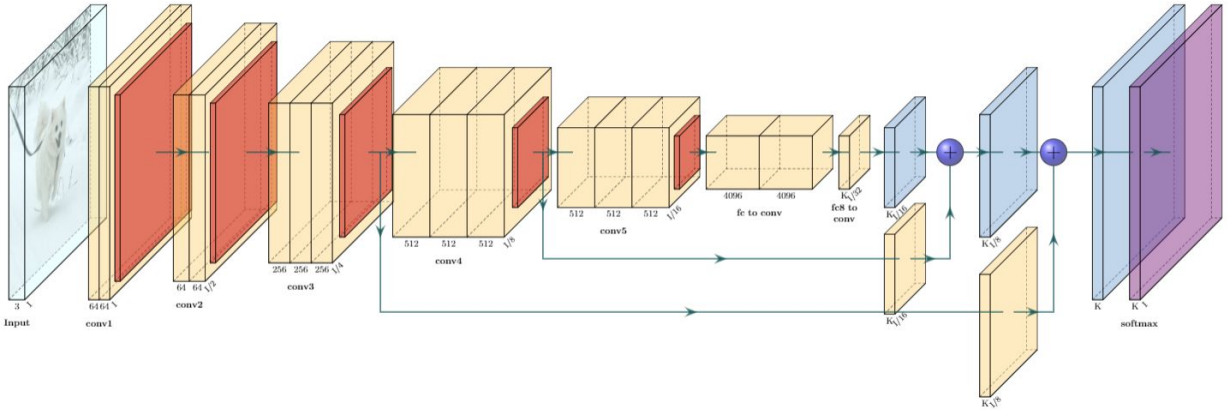
- Upsampling via Convolution
  - Generates an output feature map that is larger than the input using a transposed convolution process
  - This process is commonly called a deconvolution, but it is distinct from an actual deconvolutional process
- For s = stride, p = padding, and k = kernel size
- Input is the **resulting feature map (fm)** of convolution on the original image
- Transposed convolution is used to inflate the fm to its original dimension - where pixels are grouped into objects



Reference: https://towardsdatascience.com/types-of-convolutions-in-deep-learning-717013397f4d

Semantic Segmentation - Fully Convolutional Networks

- Fully Convolutional Networks (FCN) by Long, Shelhamer, and Darrel
  - Achieved a score of ***62.7% mean IoU on PASCAL-VOC2011*** **and** ***62.2% IoU on VOC2012***, ~10% improvement over previous best on both datasets
  - Uses downsampling to refine image features followed by upsampling to recover location information



References:
J. Long, E. Shelhamer, T. Darrell “Fully Convolutional Networks for Semantic Segmentation” CVPR (2015)

Semantic Segmentation - Benchmarks

- Pascal-VOC
  - 2011
    - 20 object classes
    - 11,530 images in train and validation set
    - 5,034 segmentations
  - 2012
    - 6,929 segmentations
- COCO
  - 500,000 object segmentations
- Cityscapes
  - 20 classes
  - 5000 images with fine annotations (pixel by pixel)

IoU (intersection-over-union) = TP／(TP+FP+FN)
TP, FP, FN # true positive, false positive, and false negative **pixels**,

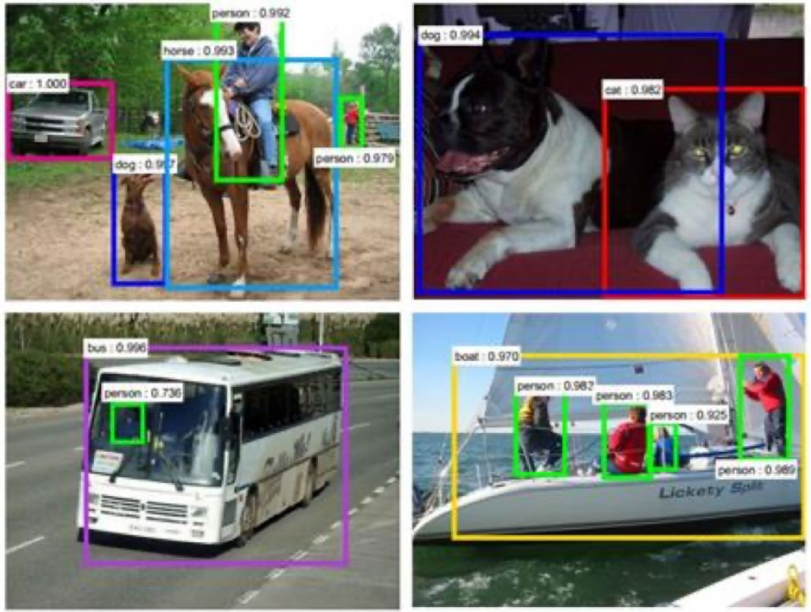
Results on the Cityscapes dataset

Method	mIOU
pre-release version of dataset	
Adelaide_Context [40]	66.4
FCN-8s [14]	65.3
DeepLab-CRF-LargeFOV-StrongWeak [58]	64.8
DeepLab-CRF-LargeFOV [38]	63.1
CRF-RNN [59]	62.5
DPN [62]	59.1
Segnet basic [100]	57.0
Segnet extended [100]	56.1
official version	
Adelaide_Context [40]	71.6
Dilation10 [76]	67.1
DPN [62]	66.8
Pixel-level Encoding [101]	64.3
DeepLab-CRF (ResNet-101)	70.4

Object detection and segmentation

~2012 – 2020: ConvNets dominate all vision tasks

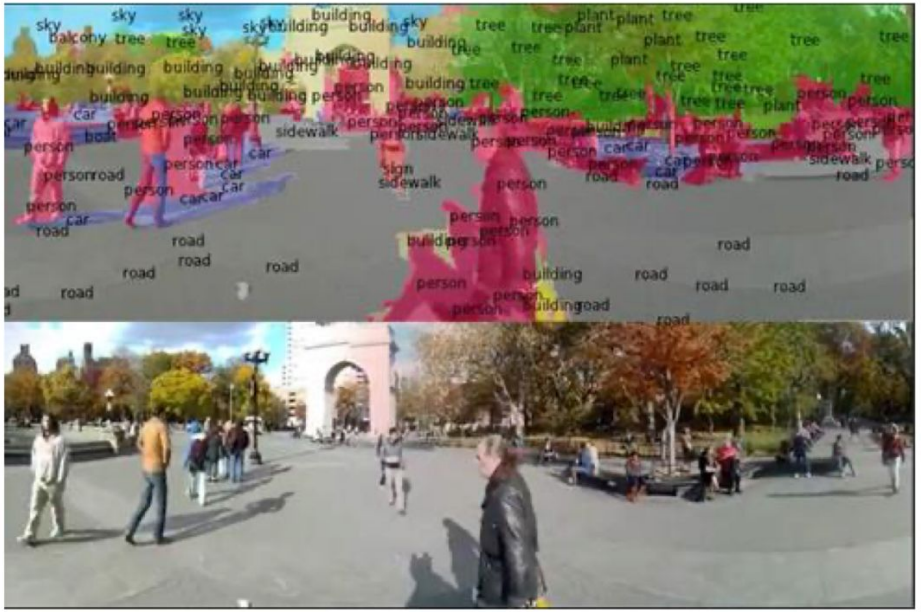
Detection



Figures copyright Shaoqing Ren, Kaiming He, Ross Girschick, Jian Sun, 2015. Reproduced with permission.

[Faster R-CNN: Ren, He, Girshick, Sun 2015]

Detection

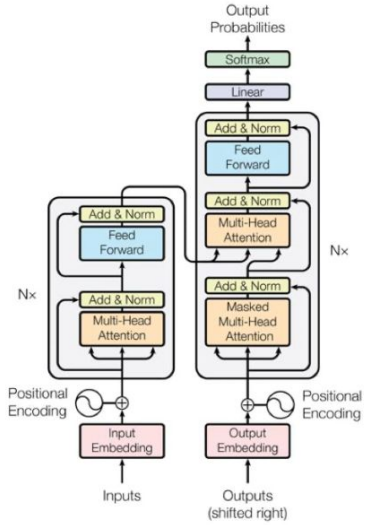


Figures copyright Clement Farabet, 2012. Reproduced with permission.

[Farabet et al., 2012]

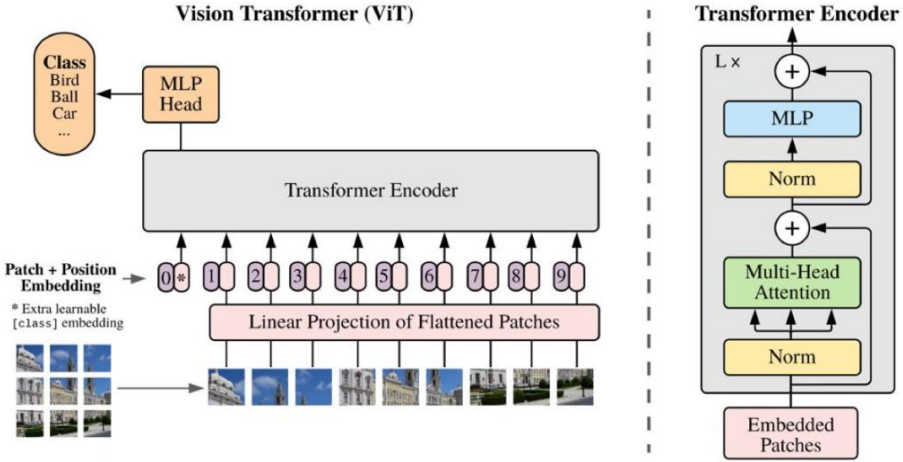
2021 - Present: Transformers have taken over

2017: Transformers for language tasks



Vaswani et al, “Attention is all you need”, NeurIPS 2017

2021: Transformers for vision tasks



Dosovitskiy et al, “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale”, ICLR 2021

What happens now? - Segment Anything (SAM 1 and 2)

Let's try the demos together!

https://segment-anything.com/

https://ai.meta.com/sam2/



A closer look at the SAMs

Promptable Segmentation

- Unlike traditional segmentation models, **SAM is promptable**—it generates segmentation masks based on input prompts.
- **Types of prompts supported:**
  - **Points** (positive or negative clicks)
  - **Boxes**
  - **Masks**
- This design makes SAM highly interactive and versatile across applications.

A closer look at the SAMs

- **Image Encoder**: A powerful **Vision Transformer (ViT-H)** extracts rich image features.
- **Prompt Encoder**: Converts user inputs (points, boxes, masks) into learnable embeddings.
- **Mask Decoder**: A lightweight **transformer-based decoder** combines image and prompt embeddings to generate segmentation masks.

This is an example of a **multi-modal fusion architecture**, combining vision and prompt information.

A closer look at the SAMs

SAM was trained on **SA-1B**, a **massive segmentation dataset** released by Meta AI:

- **Over 1 billion masks**
- **11 million images**
- Covers a wide range of object categories and contexts

It enables robust learning for general-purpose segmentation.

Zero-shot Generalization

- SAM is designed as a **foundation model for segmentation** (similar to how CLIP is for vision-language).
- It performs well **out of the box** on unseen images, objects, and domains **without fine-tuning**.
- Easily adaptable to downstream tasks in medicine, satellite imagery, video, and more.

Today’s agenda

- A brief history of computer vision and deep learning
- Visual perception and understanding: What makes a good visual representation? *(Lab to follow after the lecture!)*
- Classic image processing techniques (a quick touch and recap)
- Image classification and beyond
- Object detection and segmentation
- Other more advanced topics on visual and multimodal generation

Tasks Beyond Image Classification

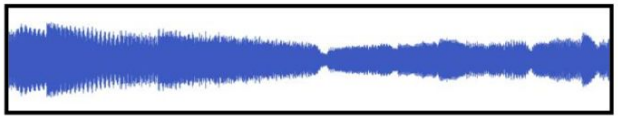
Video Classification



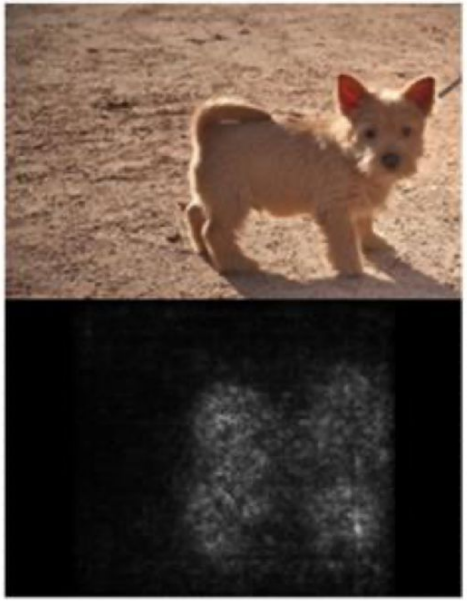
Running? Jumping?

Multimodal Video Understanding



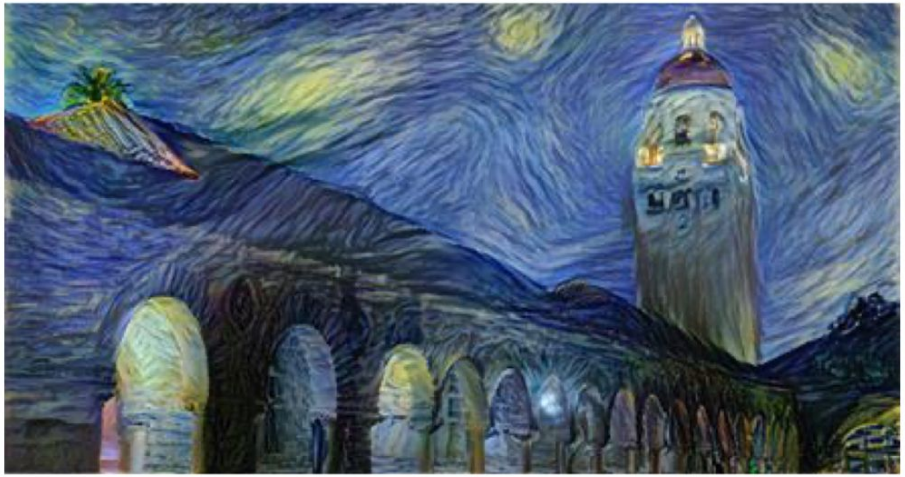


Visualization & Understanding



Beyond 2D Recognition: Generative Modeling





Style Transfer

Beyond 2D Recognition: Generative Modeling



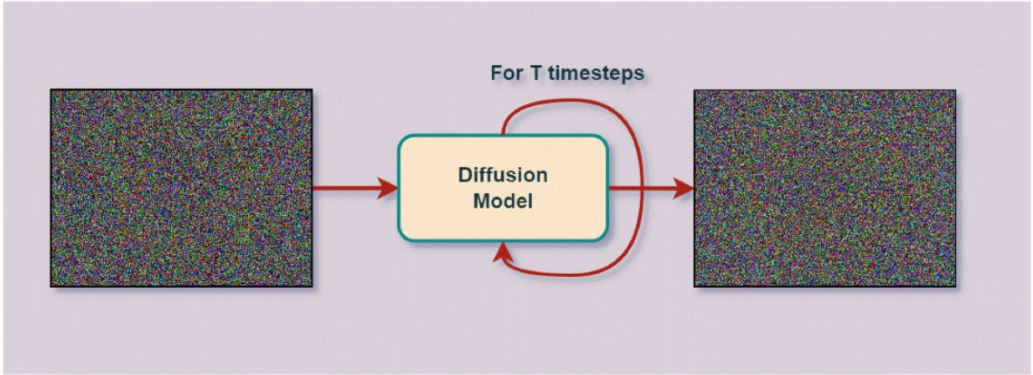
This image is public domain

"Teddy bears working on new AI research underwater with 1990s technology"

DALL-E 2

Beyond 2D Recognition: Generative Modeling

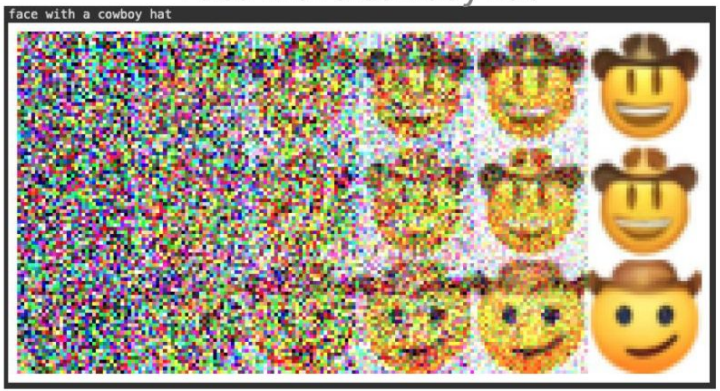
Image Generation using Diffusion Models



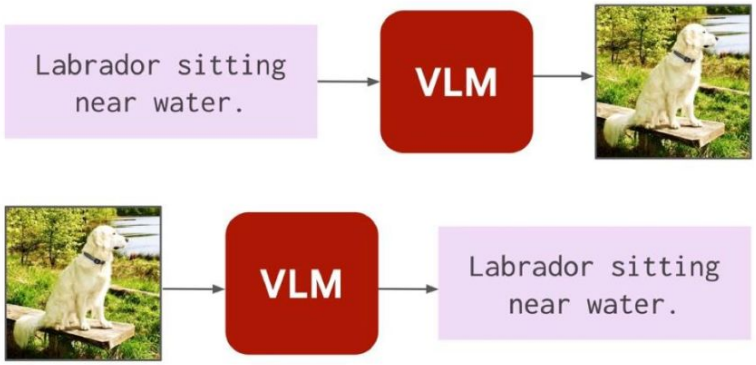
https://learnopencv.com/image-generation-using-diffusion-models/

You will learn and implement a generative model in Assignment 3 that generates emojis from text inputs

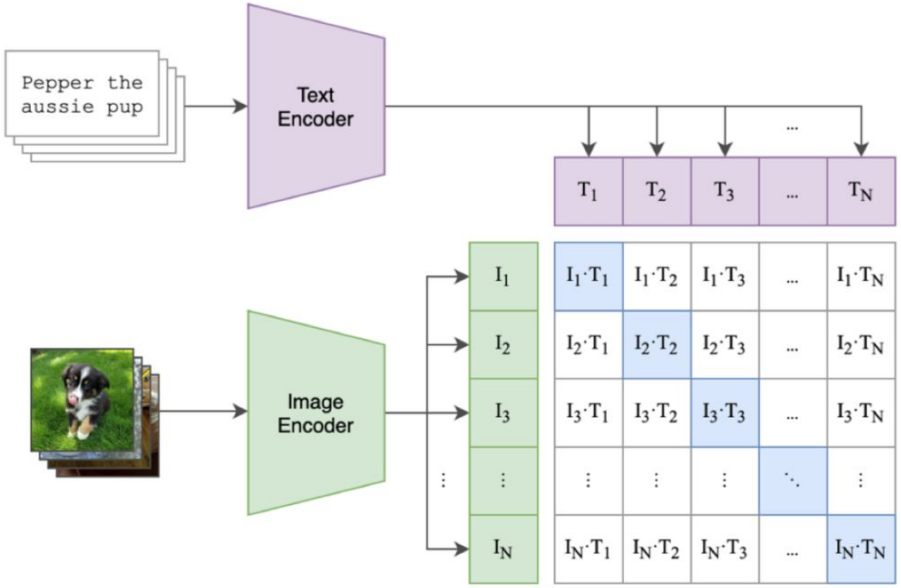
Face with a cowboy hat



Beyond 2D Recognition: Vision Language Models



Yasunaga, Michihiro, et al. "Retrieval-augmented multimodal language modeling." arXiv preprint arXiv:2211.12561 (2022).



Contrastive pre-training in CLIP. The blue squares are the pairs for which we want to optimize the similarity. Image derived from https://github.com/openai/CLIP

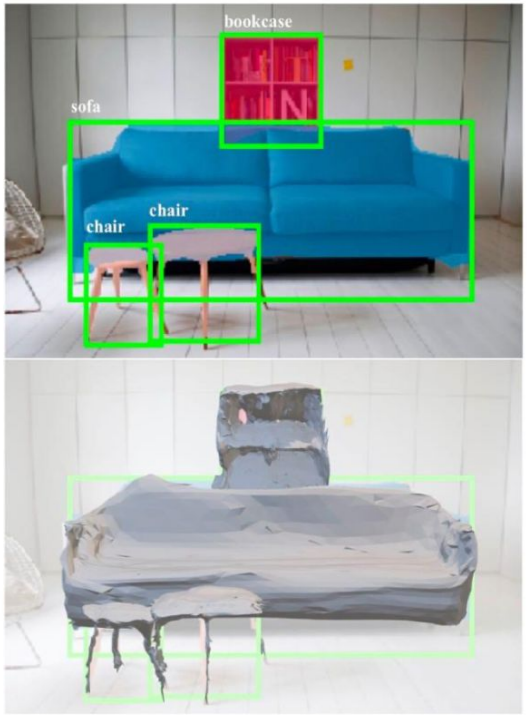
Beyond 2D Recognition: 3D Vision



Choy et al., 3D-R2N2: Recurrent Reconstruction Neural Network (2016)



Zhou et al., 3D Shape Generation and Completion through Point-Voxel Diffusion (2021)



Gkioxari et al., "Mesh R-CNN", ICCV 2019

What makes a good visual representation again?

See you in the lab session!